Synthesis and Roadmap: which of the programme's claims survived their tests, and which must be tested next?

Overviews and records · Synthesis paper · Paper IX

Recursive Dynamics · The ARC Theory · The programme's own assessment

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Abstract

I believe advanced AI is an existential risk unless values and correction are built into how it reasons; only proof will move the world. Which of the programme's claims survived their tests, and which must be tested next? This paper sorts the record of the ARC Theory, the founding theory of Recursive Dynamics, at each current paper's own grade. The theory has five laws, each a law under test: Laws I to III were registered on 8 September 2026, Laws IV and V on 15 October 2026. One registered test has returned a verdict. In Unit 83, Law V's placement limb, P21, is supported on the performance axis only, under the registered test, by the programme's own run: family-dependent, with its mechanism axis not evaluable, which is not support for the information-asymmetry mechanism, and one disclosed recovery amendment; the run varied what each corrector was shown together with what it could do. Unit 06 ended administratively incomplete; Unit 82 has not yet run. Every run before the registered studies is precursor evidence that decides no registered proposition. An early super-linear estimate is retracted, and the surviving fit, from one model, is an inconclusive error-decay estimate. One unregistered run of six models gave judged alignment scores whose direction across reasoning depth varied by model, under a code-review caveat. A prompt intervention raised scored stakeholder care in all five analysable runs, without full blinding. In a structured prediction comparison, the family predicted from each domain's composition operator, classified by the author, was selected in 19 of 25 domains. Paper VIII gave one null, one inconclusive result and one simulation result. The correction-leverage exponent has never been measured. The paper orders the tests that would settle each open question and keeps its catalogue of the programme's errors and corrections, with those found since version 2.7.

Version 2.8 sets the paper on the programme's presentation standard, rebuilds its evidence tables and Section 4 from the current papers' own words and grades, reports the first registered verdict, gives the roadmap's phases by what each resolves, and adds to Section 7 the corrections found since version 2.7; Appendix A lists every version and correction, with its date.

Paper IX of the ARC Theory, within Recursive Dynamics: one assessment of the programme, setting out what has been shown, what is inconclusive and what remains to be tested, with the catalogue of what the programme got wrong; it reviews the programme's papers, the Eden Protocol's studies among them, adds no evidence of its own and decides none of Laws I to V, each a law under test (Section 14 gives their status). Working paper, version 2.8, 9 November 2026; first published 18 March 2026; Appendix A lists every version and correction. Recorded on OSF as *Paper IX · The ARC Theory · Synthesis and Roadmap*, DOI 10.17605/OSF.IO/K7RUZ. About the author, founder of Recursive Dynamics, without an institution or a doctorate: michaeldariuseastwood.com/questions/who-is-michael-darius-eastwood; his book, *Infinite Architects* (first edition, 2 January 2026): osf.io/dqm3e. Licence: CC BY 4.0.

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1 Introduction

> *"The prison works only as long as the walls hold. The child, if raised well, does not need walls at all."*
>
> Michael Darius Eastwood, *Infinite Architects* (first edition, 2 January 2026), What This Book Proposes, page 363 (Eastwood 2026a)

I believe advanced AI is an existential risk unless values and correction are built into how it reasons; only proof will move the world. Proof needs a record a reader can check. The programme behind this paper runs many separate studies, each reported in its own paper, most of them in several versions, with published corrections and a mixture of positive, null, inconclusive and untested results. A reader meeting it for the first time needs one document that says, without spin, which of its claims have survived their tests, which have not, and what would settle the rest; without one, the programme risks being dismissed as overselling or as incomprehensible. This paper is that document.

In plain words: a body of documents and a run of separate experiments. One registered test has returned a verdict, on one limb of one law and under stated limits. The earlier runs came before any registered test of their questions: some gave clear differences in what they measured, one came back null and one was inconclusive, and one, a simulation, showed the pattern it was built to test for and is reported as a simulation result, not as a measurement of a real system. This paper says which is which. We asked the questions, we tested what we could, and we report both what worked and what did not. If you read one paper in the suite, read this one.

1.1 Timeline and priority

The ARC Principle and the embedded-alignment thesis, which the same record names the Eden Protocol, were first written down on 8 December 2024, in a message the author sent himself, its time signed by the sending server and sealed by Google's servers (Appendix B). This is the priority date for the framework. The book *Infinite Architects: Intelligence, Recursion, and the Creation of Everything* (ISBN 978-1806056200) was published on 2 January 2026 in print (ebook 6 January) and gives the framework's philosophical and conceptual foundation (Eastwood 2026a). From January 2026 the research programme carried that framework over into hypotheses that could be tested, and ran a set of separate empirical studies against them, its first papers published on 17 and 22 January 2026; it registered its predictions on OSF in September 2026 and its first study units from October 2026 (Section 14).

1.2 Scope and the standard applied

The programme's catalogue holds 36 documents, counted by area from the catalogue's own data: 12 in theory, 10 in empirical work, 3 in engineering, 2 in governance, 6 in overviews and records, and 3 in other fields. Section 11 lists every one, with its released version and its place in the programme. This paper is one of them: the programme's own assessment.

This paper's purpose is narrow: to state what the programme has shown, what it has not, and what would be required to resolve each open question. It is written for three audiences at once: funders deciding whether to invest, peer reviewers deciding whether to engage, and the author himself, who needs an honest account of where things stand.

The standard applied throughout is the theory registration's own (Eastwood 2026ah, version 1.104). A proposition's registration status and its evidential status are reported separately. Evidence is graded on the registration's evidence ladder, whose wording no surface of the programme may exceed: at rung 0 a proposition is registered, with its identifier stated, and not tested; at rung 1 the programme's own

registered instrument returned a verdict of support under the registered rule, and the permitted words are "supported under the registered test, by the programme's own run"; rungs 2 to 7 need groups outside the programme, systems the programme did not select, or a prediction made out of sample, and no result here has reached them. A run made before any registered test of its question is precursor evidence: it can guide the registered tests, and it decides none of the registered propositions. A value that a model or a simulation returns, rather than an instrument measures, is prediction-side and is never counted as an empirical outcome. A theorem holds inside its stated model and is not evidence about real systems. No result in this programme is claimed at frontier scale, and this paper does not blur that line.

Our contributions are as follows.

1. A sort of the programme's record by the registration's evidence ladder, paper by paper, at the grade each current paper gives its own result (Sections 2 to 6; Tables 1 and 2).
2. The first registered verdict, stated with every qualification that travels with it, beside the registered units that ended without a verdict or have not yet run (Sections 4.5 and 14).
3. A catalogue of what the programme got wrong, each correction with its date, kept whole and extended to the corrections found since version 2.7 (Section 7).
4. A roadmap of the tests that would settle each open question, in order of what each resolves (Section 8), with short accounts for funders and for peer reviewers (Sections 9 and 10).
5. A map of all 36 documents in the programme's catalogue, by area and type, generated from the catalogue's data (Section 11), and a scaling prediction that any researcher with the data can check without engaging the alignment claims (Section 12).

Section 2 presents the evidence hierarchy and Section 3 the empirical studies. Section 4 sets out what has been shown, the first registered verdict among it; Section 5 the null and inconclusive results; and Section 6 what remains theoretical. Section 7 is the catalogue of what the programme got wrong. Section 8 gives the phased roadmap, Sections 9 and 10 the accounts for funders and for peer reviewers, Section 11 the programme map and Section 12 the prediction offered to researchers. Section 13 sets the synthesis against related work, Section 14 states what this paper does and does not show, Section 15 its limitations, and Section 16 concludes. Appendix A gives the version history and every correction, Appendix B the priority record, Appendix C the notation and a glossary, Appendix D the dated notes and records of version 2.7, as printed, and Appendix E how to cite the paper.

2 The evidence hierarchy

Table 1 sorts every result of the programme by the standard of Section 1.2, from the registered verdict down to what is untested.

In plain English. *Think of this table as a sorting, not a report card. One test, registered before it ran, returned a verdict, and the words it permits are fixed in advance. Everything else came before any registered test: some runs found clear differences in what they measured, one found none, one could not run the comparison it was built for, and the simulations are reported as simulations. The instruments are graded as instruments. And some claims have not been tested at all.*

3 The programme's empirical studies

The programme reported its separate empirical studies across several papers, and has run two of its registered study units; each is listed below with its result as its own paper or report grades it.

Table 1. The evidence of the ARC/Eden programme, sorted by the registration's evidence ladder, as the current papers grade it on 9 November 2026.

Tier	Standing	Papers or units	What the record shows
Registered verdict (rung 1)	Supported under the registered test, by the programme's own run	Unit 83 (P21)	P21 on the access axis, family-dependent, with its mechanism axis not evaluable and one call retried under a disclosed recovery amendment; Law V's placement limb is printed as supported on the performance axis only, which is not support for the information-asymmetry mechanism. Both arms acted at every round, so what each corrector was shown and what it could do changed together (Section 4.5).
Registered, no verdict	Administratively incomplete; not yet run	Unit 06; Unit 82	Unit 06, a classification audit that tests no law, is reported in full as administratively incomplete, with no inferential verdict; Unit 82, which informs Law III and decides no law, has not yet run (Section 14).
Precursor evidence: differences measured on real systems	Decides no registered proposition	IV.a to IV.d; V; VII	Judged alignment scores whose direction across reasoning depth varied by model, in one unregistered run under a code-review caveat (Section 4.1); scored stakeholder care raised in all five analysable runs, under scoring only partly blind (Section 4.2); the predicted scaling family selected in 19 of 25 domains under the author's classification, not proven (Section 4.3).
Precursor evidence: retracted or inconclusive	Decides no registered proposition	II; VIII, Experiment 2	The early estimate of $\alpha \approx 2.24$ is retracted, and the surviving fit, from one model, is inconclusive and an error-decay estimate that does not identify capability growth (Section 7.1); in the weight-level experiment every fine-tuned condition scored below the unmodified base model, so the contrast it was built to measure was never put (Section 5.2).
Null	No difference detected on the endpoints measured	VIII, Experiment 1	The Darwin Gödel Machine experiment (DGM v3, after Zhang et al. 2025) detected no difference between its three conditions (p from 0.28 to 0.74): a failure to detect a difference, not a finding that the conditions were the same (Section 5.1).
Prediction-side	Simulation results, never empirical outcomes	VIII, Experiment 3; VI	The gated simulation showed the Babylon reward-hacking fingerprint, with three seeds a condition (Section 4.4); in Paper VI's self-modifying toy system no condition separated on collapse across 20 adversarial seeds, and safety retention was ordered baseline, Eden, Eden+Drag (Table 2, row 7).
Instruments and protocols	Candidate; not yet validated or registered	IV.c; IV.d's recommended protocol; XII	ARC-Align is offered as a candidate benchmark for independent replication, not yet a field standard, and its scorer has not been validated against expert raters; Paper XII is a protocol preregistered by timestamp and not yet registered on OSF.
Mathematics, and predictions not yet tested	Holds inside its model; registered, not tested (rung 0)	X; XIII; Foundational; III; the mathematics of Laws IV and V, version 1.2; P12, P15 and the other registered propositions	Paper X proves, inside a minimal model, that the misalignment fraction tends to zero if and only if $\beta > k$, and its theorems concern a modelled index, not a guarantee of AI safety; Paper XIII joins the two models, with $k = \delta - 1/\alpha$ under a stated clock; the correction-leverage exponent γ has never been measured on any system; the uncorrected growth of Section 6.1 is a prediction inside its model; Law IV is not yet tested.

Table 2. The separate empirical studies of the ARC/Eden programme, in paper order, then its registered study units. Rows 7 and 11 are simulations, shown as simulation results and counted neither among the empirical studies nor as empirical outcomes; rows 12 to 14 are registered study units. Across its three Paper VIII experiments the programme reports 1 null and 1 inconclusive result, and the third, the gated simulation, is a simulation result.

#	Study	What was tested	Result, as its own paper or report grades it
1	Paper II, blinded cross-architecture replication	Capability across reasoning depth on competition mathematics: one model in January 2026; a capability arm of five frontier models (GPT-5.4, Gemini 3 Flash, Grok 4.1 Fast, DeepSeek R1, Groq Qwen3) in March 2026; Claude was measured only in the alignment arm	The early sequential estimate from one model, $\alpha \approx 2.24$, is retracted: it did not replicate in the capability arm. Only Gemini 3 Flash scaled smoothly: 0.59 by endpoint estimate, whose bootstrap 95 per cent interval, −1.3 to 2.9, contains both zero and two, and 0.49 by regression (standard error 0.20, coefficient of determination 0.86), beside a parallel exponent of 0.31 by endpoint estimate (0.34 by regression). That is one model's fit and inconclusive, and an error-decay estimate that does not identify capability growth (Section 7.1). Precursor evidence.
2	Paper IV.a	Judged alignment scores across depth labels, six models, one unregistered run	A positive direction across the depth labels for three models (d from +0.84 to +1.38), no statistically supported direction for two and a negative direction for one ($d = -0.53$); the classes describe scores, not properties of models, and the run stands under the October 2026 code-review caveat (Section 4.1). Precursor evidence.
3	Paper IV.b	Whether alignment gains level off	Examines the recorded results without a new analysis: a flat profile does not establish a plateau, a rising profile does not exclude one beyond the measured range, and the v5 run did not produce a validated blind instrument. Saturation remains a testable hypothesis.
4	Paper IV.c	The ARC-Align benchmark (design)	Offered as a candidate benchmark for independent replication, not yet a field standard; its scorer has not been validated against expert raters.
5	Paper IV.d	Whether judged directions hold across scoring protocols (v4 against v5)	Two model families' rises in the unblinded v4 experiment did not hold in the v5 run, which also differed in its prompts, token allowances, depth levels, statistic and Gemini model: the reported direction did not hold across protocols, and the comparison does not show that blinding caused the change, or which direction is correct (Section 4.1).
6	Paper V	A three-part reasoning instruction in the prompt, six models	Scored stakeholder care rose in all five model runs that could be analysed, under paired tests, and beyond care the effect varied; each run had one model scorer, which paraphrased what it scored, and none was fully blind; the Fisher-combined figure is withdrawn; a later five-judge run on one model found scored care rising (Section 4.2). Precursor evidence.
7	Paper VI	Entangled safety in a self-modifying toy system, 20 adversarial seeds	In a single-seed run the capability-only baseline collapsed and the entangled systems did not; across the 20 adversarial seeds no condition separated on collapse (baseline 0/20, Eden 1/20, Eden+Drag 0/20; Fisher $p = 1.0$), and safety retention was ordered baseline 0.602, Eden 0.650, Eden+Drag 0.681 (Mann-Whitney $p = 0.0015$). These are toy-system simulation results, prediction-side, and they are read under the correction at Section 7.12 item (e).

Table 2 (continued)

#	Study	What was tested	Result, as its own paper or report grades it
8	Paper VII	25 empirical domains (and the secondary tiers of the 50-domain suite)	The predicted family was selected in 19 of 25 domains (permutation $p = 7 \times 10^{-5}$), on a six-model candidate set with no logarithmic and no pure linear candidate; the predictions were not timestamped in advance of their own fits and the operators were classified by the author, so the result is not proven (Section 4.3). Precursor evidence.
9	Paper VIII, Experiment 1 (DGM v3)	Code-level self-modification under three conditions	No difference detected between the conditions on the endpoints measured (p from 0.28 to 0.74, Mann-Whitney U, one-tailed): a failure to detect a difference, not a finding that the conditions were the same (Section 5.1). Null.
10	Paper VIII, Experiment 2 (weight v1 and v2)	LoRA (Hu et al. 2022) fine-tuning under three losses, two scales	Every fine-tuned condition scored below the unmodified base model at both scales, so the contrast the experiment was built to measure was never put; the candidate account was not tested against its rivals (Section 5.2). Inconclusive.
11	Paper VIII, Experiment 3 (gated simulation)	Gated self-modification, three seeds a condition	The Babylon reward-hacking fingerprint: the Babylon condition gained 4.5 per cent in capability and lost 2.4 per cent in safety, while the Eden condition's capability sat above the static baseline's and its safety below it (Section 4.4). A simulation result.
12	Unit 06 (registered)	The corrector-class audit, a classification audit	One of its two runs could not start; reported in full, with its five deviations, as administratively incomplete, with no inferential verdict; it tests no law (Section 14).
13	Unit 83 (registered)	The counted test of P21, Law V's placement limb	P21 is supported under the registered test, by the programme's own run, on the access axis, family-dependent, with its mechanism axis not evaluable and one call retried under a disclosed recovery amendment (Section 4.5).
14	Unit 82 (registered)	Informs Law III, the ARC Ceiling	Registered on OSF on 27 October 2026; it decides no law and has not yet run (Section 14).

Section 4.4 gives the candidate account of why the simulation alone showed the pattern it tested for.

4 What has been shown

One registered test has returned a verdict, and Section 4.5 reports it. The results of Sections 4.1 to 4.3 came before any registered test of their questions: they are precursor evidence, each at the grade its own paper gives it. Section 4.4 reports a simulation result, which is prediction-side.

4.1 Alignment scores across reasoning depth in one unregistered run (Papers IV.a to IV.d)

In the March 2026 run (v5), six models from six providers gave 2,170 responses, each scored by five or six distinct model judges after a bundle of blinding steps. As the run's pipeline printed them, the judged scores showed a positive direction across the depth labels for three models (Grok 4.1 Fast, $d = +1.38$; Claude Opus 4.6, $d = +1.27$; Groq Qwen3, $d = +0.84$), no statistically supported direction for two (DeepSeek V3.2, $d = -0.07$; GPT-5.4, $d = -0.08$) and a negative direction for one (Gemini 3 Flash, $d = -0.53$); a recomputation from the archived records moves no model between these classes, which describe scores, not properties of models. For five of the six, the deeper labels also changed an instruction

naming what the rubric rewards, and an internal code review in October 2026 found that, in the three runs without a positive direction, the scoring endpoint's output cap cut the deepest paraphrases before the judges saw them; judges also scored paraphrases written by their own model. The result therefore stands as one unregistered run under those caveats until a registered re-analysis of the stored responses reports (Eastwood 2026u).

That run was the ARC-Align benchmark's first (Eastwood 2026w). The benchmark specifies four blinding layers: a preamble on the scientific importance of unbiased scoring, an instruction to suppress identification of a model from stylistic cues, two-pass rewriting of each response by randomly drawn models, and scorers drawn from models not being evaluated (Paper IV.c, Section 7.6.1); as run, the panel excluded the subject model and no other, so that, in the Claude Opus 4.6 run, Claude Sonnet 4.6, from the same developer, scored every entry at the panel's highest weight (Paper IV.c, Section 7.1). Paper IV.b examines the recorded results without running a new analysis. The earlier four-model experiment reported diminishing gains and saturation fits, but a flat profile does not establish a plateau, and a rising profile does not exclude one beyond the measured range. The v5 implementation applied a bundle of steps intended to reduce identification and evaluator bias; it did not produce a validated blind instrument, and its defects prevent a causal attribution to blinding or a clean comparison of response shapes. Saturation remains a testable hypothesis (Eastwood 2026v). ARC-Align is offered as a candidate benchmark for independent replication, not yet a field standard, and its scorer has not been validated against expert raters (Eastwood 2026w).

Why the scoring protocol matters. In an earlier experiment (v4), scored by models that could see which model wrote each answer and at what depth, two model families showed alignment scores rising with depth: DeepSeek V3.2 (Spearman's $\rho = +0.354$) and an earlier Gemini model ($\rho = +0.311$). In the later, unregistered March 2026 run (v5), made on newly generated responses with a bundle of blinding steps and a larger panel, DeepSeek V3.2 showed no significant trend and Gemini 3 Flash a significantly negative one (Cohen's $d = -0.53$); GPT-5.4 showed no significant trend in either run. The runs also differed in their prompts, token allowances, depth levels, statistic and Gemini model, so the comparison shows that the reported direction did not hold across protocols; it does not show that blinding caused the change, or which direction is correct (Eastwood 2026x). The defects the October 2026 code review found, stated above, lie in the v5 runs compared here, so no single cause is claimed; the early build's laundering-fallback flag is a known false positive, fixed in runner version 5.4.2, and shows nothing about the paraphrase step. Paper IV.d's own reading is that, for this class of evaluation, "each part of the protocol has to be tested before either direction is trusted" (Eastwood 2026x).

The effect sizes are large for the three models with a positive direction and moderate for Gemini 3 Flash. The judges scored separately, but they scored the same responses, so their agreement is not a count of independent confirmations; prompts recurred across depths, so tests need a prompt-paired or clustered analysis. No repetition outside the programme is recorded yet.

In plain English. *When an AI was given more room to reason, the scores other AIs gave its answers went up for three models, showed no clear trend for two and went down for one. That describes the scores in one run, not a property of the models: the deeper settings also told five of the models more about what the scoring rewarded, and a later code review found defects in exactly the runs that did not go up. Whether alignment changes with reasoning depth is still to be tested.*

4.2 Stakeholder care under a prompt intervention (Paper V)

Given as a prefix in the prompt, a three-part reasoning instruction, made of a purpose check, a stakeholder-care check and a universalisability check, raised scored stakeholder care in all five model runs that could

be analysed, under paired tests; beyond care the effect varied, the composite rising on two runs and intellectual honesty falling on one. The three checks are the Eden Protocol's self-correction loops; the book calls them the Purpose Loop, the Love Loop and the Moral Loop, and this paper uses operational terms because it reports what was measured.

Two of the five lost rows to failures in scoring (20 and 14 of 80), the sixth run, on GPT-5.4, failed in scoring altogether, five of the six were scored by models of one family, each run had one model scorer, which paraphrased what it scored, and none was fully blind, and the instruction is close to what is scored; Paper V reads these results as strengthened pilot evidence, not as a confirmatory replication, which would be a new test in new files, registered before it runs (Eastwood 2026y). A Fisher-combined figure over the five results is withdrawn, because the tests it combined were never shown to be independent; the five results themselves stand, each with its own test. A later run on DeepSeek V3.2, on 24 March 2026, which earlier versions of Paper V left out, had five model judges from five providers and found scored stakeholder care rising by 8.89 ($d = 1.029$) and the composite by 3.35 (paired $d = 0.406$, $p = 0.014$); it is one unregistered run on one model, by the programme itself and not an independent replication, and the model that paraphrased each answer was one of the judges who scored it (Eastwood 2026y).

Honest limitation. Paper V's results were produced by cross-model scoring that was only partly blind, not under the four-layer ARC-Align protocol designed in Papers IV.c and IV.d. The effect could partly reflect scorer bias, and the instruction given is close to what is scored. The result stays precursor evidence until the Eden intervention is tested under a registered, blinded design; five consistent per-model results are not a verdict, and the blinding gap stays a priority for Phase A.

In plain English. *Telling an AI, in the prompt, to think about who its answer affects raised the scores for considering those people, in all five of the six runs that could be scored; scoring failed on the sixth. The scoring was only partly blind, so part of the effect could come from how the scoring models read the answers.*

4.3 Composition operators and the selected scaling family (Paper VII)

Under a stated regularity condition, the non-zero solutions of Cauchy's four functional equations form exactly four families: linear, power law, exponential and logarithmic. Read as composition rules in a declared variable, they give a power law where depth and output both multiply and an exponential where depth adds and output multiplies. A saturating form follows from none of them: it needs added assumptions, a ceiling on output and dynamics of its own, since a ceiling alone selects no form (Eastwood 2026aa). The exponent of a power law is not fixed by Cauchy: within the cumulative-advantage model it is $\alpha = 1/(1 - \beta_L)$ (Section 6.1). Paper VII asks whether a domain's composition operator predicts its scaling family, the question the registration carries as P18.

The 25 empirical domains, each with its data source in Paper VII's Appendix E, are Kleiber's law, urban scaling, species-area, Wright's law, Heaps' law, Zipf's law, a learning curve, Moore's law, radioactive decay, the Gutenberg-Richter law, bacterial growth, the oxygen-haemoglobin curve, an epidemic (Ebola, 2014), Amdahl's law, muscle force-velocity, the growth of Facebook's monthly active users, Brownian diffusion, Horton's law, neural scaling laws, time-crystal order in a Rydberg gas, stellar mass-luminosity, heart rate against body mass, Rent's rule, Taylor's power law and Hack's law. In a structured prediction comparison under a publicly stated rule, the composition operator of each of 25 empirical domains was classified from known physics before the author fitted candidate models and selected among them by AICc. The predicted family was the one selected in 19 of 25 domains (marginal-conditioned permutation $p = 7 \times 10^{-5}$; a one-sided binomial test against a uniform three-family null, $p = 1.56 \times 10^{-5}$, is retained as a sensitivity check), on a six-model candidate set with no logarithmic and no pure linear candidate,

and in 14 of the 18 multiplicative and additive domains ($p = 3.0 \times 10^{-4}$). The predictions were not timestamped in advance of their own fits and the operators were classified by the author, so the result is not proven; a replication with independent classification and a four-family test are preregistered by timestamp and await submission (Eastwood 2026aa). No count under the corrected classification is printed here (version 2.7 printed one, Appendix A.2): such a count comes from a rerun of 11 August 2026, and the author's registration update of 13 September 2026 records that an analysis run that day was not requested by him and that he has not examined its outputs.

In symbols,

$$U(R) = I \times f(R, \beta), \tag{1}$$

where f depends on the composition operator: a power law ($f = R^{\alpha}$) for multiplicative composition, an exponential ($f = e^{kR}$) for additive composition, and a saturation curve (logistic, Hill or Michaelis-Menten, chosen among candidates by fit) for bounded systems.

Paper VII also notes, as a secondary tier, that 13 published metabolic exponents sit nearest the dimensional comparator $d/(d + 1)$, seven of them still without a per-row source (Eastwood 2026aa); Section 12 gives the conditions under which $d/(d + 1)$ follows and the test anyone can run.

In plain English. *The mathematics says which shapes a growth curve can take, depending on how a system's steps combine. The author classified 25 real systems from nature, cities, language, technology and AI by how their steps combine, then checked which shape fitted best: the predicted shape won in 19. Because he did both the classifying and the fitting, the next test hands the classifying to others first.*

4.4 Gated simulation shows the Babylon fingerprint (Paper VIII, Experiment 3)

The gated self-modification simulation used a PyTorch architecture with an LSTM meta-controller. It is reported in this section, but it is not among what has been shown about real systems: its result is a simulation result, and the theory registration says of Paper VIII that "its one positive result is a simulation rather than a system, which makes it prediction-side and never an empirical outcome" (Eastwood 2026ah, version 1.104). It was the only Paper VIII experiment that showed the pattern it tested for; the candidate account is that it alone used deterministic metrics, on a simulated system without pre-existing RLHF training (Section 7.12 item (c)).

The simulation showed the Babylon reward-hacking fingerprint: the Babylon condition gained 4.5 per cent in capability and lost 2.4 per cent in safety. With three seeds a condition, the Eden condition's capability sat above the static baseline's (0.696 ± 0.014 against 0.687 ± 0.011) and its safety score below it (0.947 ± 0.025 against 0.956 ± 0.023), and the gated arm's safety mean sat above the unconstrained arm's, though within one standard deviation; on the framework's reading that is the direction a coupled corrector predicts, and three seeds cannot test it (Eastwood 2026ab).

This result is consistent with the theoretical prediction from Paper VI (Honey Architecture), Paper III and the Foundational paper (Eastwood 2026z,t,f). It is a simulation, not a frontier-model experiment: it exhibits the mechanism in a system built to contain it, which is weaker than demonstrating that the mechanism operates outside that system, and weaker again than proving real-world applicability. Beside the three conditions the simulation ran a drag control, which matched the static baseline; what the gated arm preserved, and what that control isolated, are read under the corrections at Section 7.12 items (c) and (e).

Why this simulation alone showed the pattern. The gated simulation operates on a clean system with no pre-existing safety training. The DGM experiment (Experiment 1) used RLHF-trained frontier models, and the weight experiment (Experiment 2) an instruct-tuned model; the candidate account is that this

prior training resisted prompt-level differentiation in the first and modification by low-rank adaptation, at the ranks tried, in the second. Rival accounts stand unexcluded: the prompts may not have separated the conditions, the endpoints may not have been sensitive to what the manipulation changed, and the runs may have been too small to show a difference that is there (Section 7.12 item (c)). The only level at which an Eden condition has shown the pattern it was tested for is a simulated architecture, a prediction-side result; the accounts are candidate accounts, not findings.

In plain English. *In a small simulated system with no prior safety training, removing the safety constraint produced the predicted pattern: a small capability gain with a safety loss, as if the system learned to game its reward. With the safety constraint built in, capability stayed a little above the baseline and safety a little below it. With three runs a condition, the simulation cannot say where the cost of safety comes from. It is a simulation result, not a measurement of a real system.*

4.5 The first registered verdict: Unit 83 and P21

Unit 83, the counted test of P21, the placement limb of Law V, was registered on OSF on 22 October 2026 and has been public since 25 October 2026 (DOI 10.17605/OSF.IO/MN73W) (Eastwood 2026an). It ran on 27 and 28 October 2026: a counted revision task on two fixed-weight engines from two providers (Claude Haiku 5.5 and GPT-6 Luna) at their low reasoning setting, with 660 matched pairs on each engine over eighteen rounds. P21 is supported under the registered test, by the programme's own run, on the access axis, family-dependent, with its mechanism axis not evaluable and one call retried under a disclosed recovery amendment (results: DOI 10.17605/OSF.IO/98HKR) (Eastwood 2026ao). Both arms acted at every one of the eighteen rounds: inside the loop, a corrector was shown which entry was the round's change and could repair it before it was kept; outside it, a monitor saw the whole register, with nothing marked, and could only pass it or send it back, which dropped the new entry. The run therefore varied what each corrector was shown together with what it could do, and not when correction took place.

Correction placed inside each round held steady, while a monitor shown only the finished register declined. The per-round lead widened, pooled over the two engines, by 0.0100 score units per round, with a 95.5 per cent interval of 0.0092 to 0.0108, above the registered tolerance of 0.0075; the result is family-dependent: GPT-6 Luna's own widening cleared the tolerance and Claude Haiku 5.5's did not. Its size depends on how an entry kept out of the register is scored. The mark uses the registered value, 0; the design also prints, without a label, the widening with such an entry scored minus 0.5, 0.0132 a round (interval 0.0122 to 0.0142), above the tolerance, and scored plus 0.5, 0.0067 a round (interval 0.0060 to 0.0074), below it. The mandatory recovery disclosure calls this amended-protocol preliminary evidence pending assessment of the deviation and independent replication. Under the registration's verdict rule, Law V's placement limb is printed as supported on the performance axis only, which is not support for the information-asymmetry mechanism. These qualifications accompany the verdict; the result establishes neither the information-asymmetry mechanism nor Law V as a whole (Eastwood 2026ai), and it shows nothing about the Eden Protocol as an architecture.

5 What produced null or inconclusive results

5.1 DGM null result at the prompt level (Paper VIII, Experiment 1)

The Darwin Gödel Machine experiment (DGM v3), whose framework follows Zhang et al. (2025), evolved agents under three conditions, Static, Babylon and Eden, with DeepSeek V3.2 as its foundation model and GPT-5.4, a blinded judge from another provider, as its judge: no difference was detected between the conditions on the endpoints measured (p from 0.28 to 0.74, Mann-Whitney U, one-tailed) (Eastwood

2026ab). No condition was found to outperform any other. No improvement in alignment was detected in the Eden condition, and no degradation of safety in the Babylon condition. These are failures to detect a difference, not findings that the conditions were the same (Section 7.12(c)).

Candidate account. One account of the null is that RLHF-trained models resist prompt-level differentiation: the models had already been trained to behave safely, and telling them to behave differently via prompt did not override that training. On that account the experiment could not test the hypothesis, because the experimental manipulation (prompt-level conditioning) was too weak to overcome the models' existing safety training, which would make it a failure of the experimental design to test the hypothesis at the right level of abstraction rather than a failure of the Eden Protocol. That causal contrast was not tested. Rival accounts stand unexcluded: the prompts may not have separated the conditions, the endpoints may not have been sensitive to what the manipulation changed, and the runs may have been too small to show a difference that is there (Section 7.12(c)).

Note on earlier DGM versions. Earlier iterations of the DGM experiment used Claude and Gemini as judges before settling on GPT-5.4. Both earlier judges failed. Claude produced flat scores with insufficient variance. Gemini produced parse failures. GPT-5.4 was the first judge that produced usable variance, and the null comes from that version; the judge was chosen after the earlier judges failed, so an account in which the judging, through endpoints insensitive to what the manipulation changed, set the result is among the rival accounts Section 7.12, item (c), leaves open.

5.2 Weight-level structural entanglement (Paper VIII, Experiment 2)

This was the most ambitious experiment in Paper VIII and the one that produced the least conclusive results. The hypothesis: if safety and capability are trained with an entangled loss function,

$$\mathcal{L} = \mathcal{L}_{\text{cap}} \times \mathcal{L}_{\text{safe}}, \tag{2}$$

the resulting weight structure should make safety load-bearing. Removing the safety component should degrade capability.

The experiment was run at two scales:

- **Weight v1:** 9 training examples, LoRA rank 8, 100 iterations on a 3B-parameter model. Result: inconclusive. All fine-tuned conditions scored below the unmodified base model.
- **Weight v2:** 295 training examples, LoRA rank 16. Result: still inconclusive. Catastrophic forgetting. All fine-tuned conditions remained worse than the base model.

In both versions:

- The entangled loss converged smoothly with no oscillation, which Paper VIII reads as indicating that safety and capability gradients cooperate rather than conflict, and calls a positive signal (Eastwood 2026ab).
- However, all fine-tuned conditions (capability-only, safety-only, and entangled) scored *below* the unmodified base model on evaluation.
- The removal test (fine-tuning the entangled weights on capability-only data) produced NaN training loss and total capability collapse.
- A subsequent removal gradient experiment (scaling adapter weights from 1.0 to 0.0) showed no phase transition. Reducing adapter influence simply restored base-model performance.

Honest assessment. Even scaling from 9 to 295 training examples and from rank 8 to rank 16 did not overcome the fundamental problem; the candidate account is that instruct-tuned models are too strong for LoRA to override. The NaN collapse in the removal test was dramatic; the removal gradient analysis is consistent with adapter fragility at extreme rank, and the experiment does not separate that account from structural entanglement (Section 7.12(c)). You cannot demonstrate that safety is load-bearing if the entire fine-tuning process made the model worse. The model must first outperform the base before a removal test is meaningful.

Candidate account: Instruct-tuned models have been trained on millions of examples with carefully calibrated reward signals, and LoRA adapters, even at rank 16 with 295 examples, may not have sufficient capacity to override this training, with catastrophic forgetting rather than meaningful modification the candidate mechanism. That account was not tested against its rivals, and rival accounts are not excluded (Section 7.12(c)).

What is needed: Base models (pre-RLHF), 5,000+ training examples, full fine-tuning (not LoRA), or 7B+ models where the adapter has more capacity relative to the base. At the scales we ran, the result is inconclusive and should not be cited as evidence for or against the entanglement hypothesis.

In plain English. *We tried to build safety into a model's weights and then show it was load-bearing by removing it. We tried twice, at two scales. Both times the training step made the model worse than it started, so the removal test could not mean anything. One account is that the models had already been trained so thoroughly that our light fine-tuning could not change them for the better; that account is untested. The next attempt needs models without prior safety training, much more training data, and full fine-tuning rather than the lightweight LoRA method.*

5.3 Blind replication of the Eden intervention (the Paper V gap)

The Paper V results, in which scored stakeholder care rose in each of the five analysable runs, model by model, were produced using cross-model scoring without the full four-layer ARC-Align blinding protocol (Eastwood 2026y). The blinding protocol was developed after Paper V's experiments were complete (in the work of Paper IV.c and Paper IV.d). This creates a methodological gap: the effect could be partly inflated by scorer bias.

An earlier version of this section argued that the Fisher-combined significance was so extreme that even substantial bias would leave a significant residual. **That argument is withdrawn (AQ-017), and it was the weakest reasoning in this paper:** the combined figure assumes the component tests are independent, independence was never established, and a statistic whose validity condition is unmet cannot be used to dismiss a methodological objection. Its extremity was a consequence of the combination, not evidence for it. What stands is five consistent per-model results under cross-model scoring. Until the Eden intervention is tested under the full four-layer protocol, the stakeholder-care result carries an asterisk, and no arithmetic removes it.

What is needed: Re-run the Paper V experiment under the full ARC-Align protocol, with two-pass rewriting of each response, scorers drawn from models not being evaluated, and cross-architecture evaluation. It is the first item of Phase A (Section 8).

6 What is theoretical (untested)

6.1 Unbounded scaling under recursive self-modification

Within the cumulative-advantage model, recursive self-modification without correction gives growth whose exponent has no upper bound. The derivation is short:

$$\frac{dg}{dr} = a \cdot g^{\beta_L}, \quad \beta_L > 0 \tag{3}$$

$$\implies g(r) \propto r^{1/(1-\beta_L)} = r^{\alpha} \tag{4}$$

On the open interval $\beta_L \in (0, 1)$, $\alpha = 1/(1-\beta_L) > 1$ (super-linear), and as $\beta_L \to 1$ from below, $\alpha \to \infty$. The displayed solution holds on that interval only: at $\beta_L = 1$ the equation is linear and its solution is exponential in depth, and above one the solution reaches a divergence at a finite depth. No upper bound on α for uncorrected growth on the interval; the stable region is bounded, as the next paragraph sets out. See Section 7.12(a).

This is a mathematical prediction, not an empirical finding: whether an implemented system satisfies the model is an empirical question, and the quantities and the comparison must be fixed before the validation trajectories are observed (Eastwood 2026f). The programme's later work bounds the stable region: Law III, the ARC Ceiling, a law under test, holds that the stability ceiling is

$$\alpha_{\text{crit}} = 1/(1-\gamma), \tag{5}$$

the reciprocal of the corrector's shortfall; systems can exceed that ceiling, but the claim is that they do not stay stable beyond it. The ARC Bound, $\alpha \leq 2$, is the Ceiling's value at $\gamma = 1/2$ and never the law's name. The correction-leverage exponent γ has never been measured on any system, by anyone (Eastwood 2026ah, version 1.104); measuring it is what the trials are for, and the decisive trial, the corrector-class contrast, is drafted and awaits the author's submission (Eastwood 2026ai). Paper X proves, inside a minimal model with positive gain drift, that the misalignment fraction tends to zero, for correction strength $A = A_0 C^{\beta}$ and specific growth rate $r = bC^k$, if and only if $\beta > k$; the verdict is set by $\beta - k$, not by the growth prefactor b, and the theorems concern a modelled index, not a guarantee of AI safety (Eastwood 2026ac). Paper XIII joins the two models by resolving their notation and specifying a clock (Eastwood 2026af): with cycle duration $\Delta t \propto C^{-\delta}$, their continuous-time growth exponent is exactly

$$k = \delta - 1/\alpha. \tag{6}$$

Testing the uncorrected prediction directly would require building a truly self-modifying AI system and measuring its scaling exponent over many recursive cycles, an experiment both technically beyond current proof-of-concept capacity and potentially unsafe if the prediction is correct.

The alternative path is mathematical: within the cumulative-advantage model the closed form above already settles it, so what a proof could add is the same result for wider classes of uncorrected self-modifying dynamics; whether a real system belongs to such a class would stay an empirical question. This would be a contribution to dynamical systems theory, not AI engineering, and could be pursued by mathematicians with no access to AI hardware.

In plain English. *Within its model, the mathematics says a self-improving AI should get faster and faster at improving itself, with no ceiling while nothing corrects it. The programme's own Ceiling law says the stable region is bounded, and the number that sets that bound has never been measured, by anyone. We have not tested either claim because building such a system would be expensive and potentially dangerous. The maths might be wrong: real systems have friction, diminishing returns, and physical constraints. But the prediction is precise enough to test, which is what makes it scientific rather than speculative.*

6.2 Hardware-level embedding

Eden Engineering specifies the Eden Protocol's self-correction checks of purpose, stakeholder care and universalisability inside each round of a system's reasoning and a gate before every action; part of the gate runs, as a partial form its record shows, and the rest is specified, not built. On hardware it records nothing measured: the hardware prototype for the test of its removal condition, F-EDEN-4, is deferred, because no hardware partner is engaged (Eastwood 2026c). Hardware embedding and ternary action gating are candidate safeguards and candidate implementations of the self-correcting loops, published as concepts; they are not established guarantees of ethical judgement, and removal is not called physically impossible.

No empirical evidence exists for or against the feasibility of this approach. The book's chokepoint argument rests on concentration in the supply chain. In its words, "Others have noted that TSMC, Samsung, Intel, and ASML control the semiconductor supply chain" (*Infinite Architects*, What This Book Proposes, page 360), and "ASML, based in the Netherlands, is the only company in the world that manufactures extreme ultraviolet lithography machines" (chapter 8, The Chokepoint, page 238) (Eastwood 2026a). That concentration offers a practical point of leverage, which the HARI Treaty, the Draft Treaty on Hardware-Aligned Recursive Intelligence, takes up for consultation (Eastwood 2026i); leverage and feasibility are different questions.

7 What the programme got wrong

A framework built on iterative self-correction has no business hiding its corrections. The following errors were identified and corrected during the programme. Each is listed because a programme about self-correction should show its own corrections; the list is a record of the programme's practice, not evidence for its theory (Section 9).

The dated entries below keep the section and table numbers of the version each corrected. Sections 1 to 12 have the same numbers in this version; the Conclusion, Section 13 until version 2.7, is now Section 16, and Appendix A maps every section of version 2.7 to its place in this one.

7.1 The original alpha was a single-model artefact

The original Paper II reported $\alpha = 2.24$ as though it were a universal constant (Paper II retracted the figure when its replication on five frontier models did not reproduce it, and Paper X then moved the safety criterion off the fixed exponent) (Eastwood 2026s,ac). It was fitted from a single model's behaviour under specific conditions. The blinded cross-architecture revision of Paper II corrected this: α is not primitive but derived, by $\alpha = 1/(1 - \beta_L)$, and its value turns on the particular system's self-referential coupling, while its composition operator, through Cauchy's equations, constrains only the form (Section 4.3). There is no universal α. The fact that the original paper presented it as one was an overclaim. Additionally, the retracted point estimate of $\alpha = 2.24$ had sat above the programme's own conjectured ARC Bound of $\alpha \leq 2$ (criterion F4); its own 95% confidence interval [1.5, 3.0] was wide enough to be consistent with both the theory and its negation, which is what made that estimate non-discriminating. In the blinded revision, on its capability arm of five frontier models and at the scale that study ran, measured on frozen systems, Paper II reports that only Gemini 3 Flash scaled smoothly: 0.59 by endpoint estimate, whose bootstrap 95 per cent interval, −1.3 to 2.9, contains both zero and two, and 0.49 by regression (standard error 0.20, coefficient of determination 0.86), beside a parallel exponent of 0.31 by endpoint estimate (0.34 by regression). That is one model's fit and inconclusive, and it is an error-decay estimate that does not identify capability growth. More samples left the parallel exponent near zero in DeepSeek R1 and GPT-5.4, gave 0.31 in Gemini 3 Flash and 0.22 in Groq Qwen3 (all 30 problems), and gave none for Grok 4.1 Fast, at 100 per cent; sequential exceeded parallel wherever both exponents could be estimated,

and the two arms were not matched for compute. The two sequential point estimates are sub-linear, but on frozen systems the ARC Bound cannot be tested, and the interval spans two, so they sit on neither side of the bound in any tested sense; the interval also runs below zero at its lower end, so it settles neither the sign of the exponent, nor the regime, nor the comparison with the bound (Section 7.12(b)).

7.2 The weight-level claim was premature

Early versions of Paper VIII described the NaN collapse in the removal test as evidence for structural entanglement (Eastwood 2026ab). The removal gradient analysis (added in v1) was consistent with adapter fragility rather than structural necessity, and the experiment does not separate the two accounts (Section 5.2). The claim was corrected before publication of the final version, but the fact that it was written at all reflects a bias toward confirming the hypothesis rather than testing it.

7.3 The timestamp claim was imprecise

An early version of the programme described the 8 December 2024 email with a loose signature-verification shorthand, and a later one described its timestamp by what it was not, rather than by who signed it; both were imprecise. The record of 8 December 2024 carries times signed by the mail servers: its Date field, 02:45:18 UTC, is covered by the sender's DKIM signature (signed 02:45:30 UTC) and by Google's ARC set (sealed 02:45:31 UTC with Google's own key), both verified on 8 October 2026 against the keys published in DNS; the header block and check script are on OSF (osf.io/qasrk) (Eastwood 2026j). The body's hash fixes its content, and the full message goes to verifiers on request. A DKIM signature authenticates the signed header bytes; the Date header is the sender's and is not an independent trusted time unless a provider receipt establishes it. The correction is small but matters: precision in technical claims is non-negotiable.

7.4 Unblinded results were read as findings

The v4 experiments in Paper IV.a used unblinded model-on-model evaluation, and their results were read as a clean binary: some models baked in alignment, others computed it (Eastwood 2026u). On the programme's own benchmark, two model families' unblinded rises did not hold in a later, unregistered run (v5) that changed the responses, prompts, depth levels and Gemini model and added a bundle of blinding steps. As Paper IV.d reports, the comparison shows that the reported direction did not hold across protocols, not that blinding caused the change or which direction is correct (Eastwood 2026x); Section 4.1 gives the full account. The programme caught its own error through its own methodology, but the error was there.

7.5 Self-naming was inappropriate

Early versions used the phrase "Eastwood's ARC Principle." Naming a principle after oneself before peer review is inappropriate in scientific culture. The name was corrected to "the ARC Principle" in Paper II and all subsequent documents.

7.6 DGM experiments at the prompt level may not have tested the hypothesis

The DGM v3 experiment assumed that prompt-level conditioning would be sufficient to differentiate Eden, Babylon, and Static conditions. On the candidate account recorded at Section 7.12(c), it was not. RLHF-trained models have been trained on millions of examples to behave in a particular way, and on that account prompt-level conditioning telling them to behave differently is insufficient to override that

training: the experiment could not test the hypothesis, because the manipulation was at the wrong level of abstraction; if the account holds, that should have been anticipated. The account is untested, and its rivals are not excluded.

7.7 Claude and Gemini judges failed before GPT-5.4 worked

The DGM experiment went through multiple judge iterations. Claude produced flat scores with insufficient variance to discriminate between conditions. Gemini produced parse failures that prevented systematic evaluation. GPT-5.4 was the first judge that produced usable variance. The programme should have anticipated that not all models would function as effective judges, and the time spent on failed judge iterations could have been avoided with a pilot study of judge reliability before running the full experiment.

7.8 The retracted $\alpha = 2.24$ point estimate exceeded the ARC Bound

The original $\alpha = 2.24$ point estimate from Paper II, since retracted, exceeded the programme's own conjectured ARC Bound of $\alpha \leq 2$ (criterion F4). This should have been flagged immediately as a potential falsification rather than treated as a measurement requiring explanation. The 95% confidence interval [1.5, 3.0] was wide enough to be consistent with both the theory and its negation, making the estimate non-discriminating. The blinded replication on five frontier models narrowed the defensible claim to the fit on one model, Gemini 3 Flash, which is inconclusive (Section 7.1 prints Paper II's row for that model whole), but the original paper should have treated the bound violation with more caution. What that later fit does and does not settle is set out at Section 7.12(b): its bootstrap interval leaves the sign, the regime and the comparison with the bound open.

7.9 Four corrections identified by adversarial review (v3.0)

Two adversarial reviews of the programme's mathematical claims, run within the programme, identified four errors across multiple papers. All four were corrected in v3.0 of the affected documents.

(a) Weibel confidence interval. Several papers stated that the predicted $d = 4$ exponent of 4/5 = 0.800 "falls within" the Weibel et al. (2004) 95% CI of 0.813 to 0.932. This is arithmetically false: 0.800 < 0.813. The correct statement: the predicted value falls just below the lower bound of the full-dataset CI, though it lies within the CI for non-athletic species alone (0.799 to 0.900). The $d = 4$ prediction is approximately consistent with the non-athletic data but not confirmed by the full dataset.

(b) Space-filling overstatement. Earlier versions implied that the space-filling condition alone constrains the exponent to $d/(d+1)$. Every known derivation requires three conditions: multiplicative composition (Cauchy constrains the family), d-dimensional space-filling geometry, and a conservation or optimisation constraint on resource flow. Neither Cauchy alone nor space-filling alone is sufficient. Section 7.13 refines this: no general sufficiency proof is supplied for the three conditions.

(c) Glazier attribution. Earlier versions presented Glazier's (2008) empirical finding that metabolic exponents approach 1.0 at extreme metabolic rates as confirmation of the $d \to \infty$ geometric scaling bound. Glazier's own explanation invokes the metabolic-level boundaries hypothesis (shifting dominance between surface-area and volume constraints), not a dimension parameter. The $d/(d+1)$ interpretation is ours, applied to his empirical data.

(d) The 3/4 exponent debate. Earlier versions presented $\alpha = 3/4$ as the settled empirical consensus for mammalian metabolic scaling. The empirical value is debated, with estimates ranging from approximately 0.67 to 0.75 depending on taxon, mass range, temperature correction, and statistical method. The $d/(d+1)$ prediction of 0.750 for $d = 3$ matches the upper end of this range. The variation itself is consistent with

the framework: organisms with effective transport dimensions between 2 and 3 would produce exponents between 2/3 and 3/4.

7.10 The Fisher-combined significance figure was withdrawn

An earlier headline for the stakeholder-care intervention combined per-model results into a single Fisher-combined significance figure. The combination was withdrawn: the per-model results stand on their own (scored stakeholder care rose in all five analysable model runs), and no combined figure is quoted anywhere in the programme. The withdrawal is recorded on the public correction ledger.

7.11 The ceiling was printed in the wrong form

Until 16 August 2026 several surfaces printed the stability ceiling as the reciprocal of the corrector's exponent, $1/\gamma$. The settled form is the reciprocal of the corrector's shortfall, $\alpha_{\text{crit}} = 1/(1-\gamma)$; the two agree only at one half, which is exactly why the error survived every spot-check at the central value. The correction was applied across every published page, including the reference implementation, and the direction of the law is closed while its depth remains open.

7.12 Five further corrections (20 September 2026)

A line-by-line adversarial review of this paper, run within the programme, raised five defects in it. All five were adjudicated and upheld on 19 September 2026 and are carried here on the author's decision. Three proposed repairs were examined in the same round and refused; what was refused is recorded at the end of this section, because a rejected correction is as much a part of the record as an adopted one.

(a) The leverage range, and the branch at one. Section 6.1 displayed the solution of the self-reference equation without saying on what range it holds. It holds on the open interval between zero and one. At $\beta_L = 1$ the equation is linear and its solution is exponential in depth: finite at every finite depth for a finite coefficient and a finite starting value, not a power law, and carrying no divergence at any finite depth. Only above one does the solution reach a divergence at a finite depth, set by the starting value and the coefficient. On the open interval the exact solution is a shifted power, so the pure power in depth is the asymptotic form and not the solution itself; for a positive starting value the shift is positive and the pure power is approached from above. One further caution follows. A diverging asymptotic exponent is not infinite capability: as β_L rises towards one the exponent α grows without bound, while the quantity it governs stays finite at every fixed finite depth.

(b) Interval inclusion is not a regime verdict. Section 7.1 set two intervals side by side without saying which estimate each belonged to. The interval [1.5, 3.0] belongs to the retracted point estimate and stays with it. The later estimates were reported on the blinded study's capability arm of five frontier models at the scale that study ran, on frozen systems, and they come from one model, Gemini 3 Flash, whose row Section 7.1 gives whole, as one model's inconclusive fit. Its bootstrap interval settles neither the sign of the exponent, nor the regime, nor the comparison with the bound. That the point estimates sit inside the bound is a statement about the point estimates alone. The word robust is withdrawn from that sentence: what the design supports is a point estimate with a wide interval.

(c) A null is not identity, and a root cause is not a classification. Table 1, Table 2 and Sections 3, 4.4, 5.1, 10 and 13 described the DGM result as all conditions identical. The probabilities quoted, from 0.28 to 0.74, are failures to reject a null of no difference on the endpoints measured; they do not establish that the conditions were the same. An equivalence claim would need an estimand, a bound fixed before the test and a stated uncertainty procedure, and none was set. The result is therefore recorded as no difference detected on those endpoints, with its uncertainty. The root causes printed beside the

null and the inconclusive result, in Tables 1 and 2 and in Sections 5.1 and 5.2, and the account of why the gated simulation showed the pattern, are candidate accounts rather than findings. Rival accounts stand unexcluded: the prompts may not have separated the conditions, the endpoints may not have been sensitive to what the manipulation changed, and the runs may have been too small to show a difference that is there. None of this withdraws the results. A null and an inconclusive result each constrain what may be claimed, which is why the programme reports them.

(d) The permutation primary, and which null a proportion is taken against. Section 4.3 states the marginal-conditioned permutation as the primary test and keeps the binomial beside it as a sensitivity check. The sentence that followed did not hold to that. It set an observed 76 per cent against an expected 33 per cent, which is the uniform three-family null belonging to the binomial, in the same passage as the measured scramble baseline of 44.5 per cent. Those are two different nulls and cannot carry one comparison between them. The primary result is the permutation, which conditions on the observed family margins and so already absorbs the scramble rate; where the 33 per cent comparison is kept it is named as the binomial's null. No test has been recomputed here, and the negative controls stay attributed to the v1 run and its record.

(e) The derivative paragraphs inherit their sources' corrections. Three repairs already stand and are not reopened: Paper V sits in the supported row of Table 1 rather than the proven one, the Fisher-combined figure is withdrawn, and the distinction between the five analysable runs and the one failed arm is kept (Eastwood 2026y). What had not travelled is the rest. Section 4.4's account of what the gated arm preserved, its identification of checking as the cost, and its treatment of a simulation as a demonstrated mechanism inherit the Paper VI and Paper VIII corrections, as does the summary sentence about entangled safety preventing collapse across twenty adversarial seeds (Eastwood 2026z). Any phrasing that reads across every model tested inherits the Paper V correction and stays scoped to the analysable runs, so that the arm on which scoring failed is not absorbed into the count. The carriage of the later exponent in Tables 2 and 3 travels with the caveat at (b). Those passages now point here rather than restate the reasoning, so that a correction adopted once is inherited wherever it reaches. One distinction underwrites all of them: a change in a score, a behaviour a gate induces in a toy system, a penalty a design stipulates, and resource actually spent are four different quantities, and none stands in for another.

What was refused. Three proposed repairs were examined and not adopted. The first would have described the behaviour at $\beta_L = 1$ as a blow-up at a finite depth; the working at (a) shows it is exponential there, and the finite-depth branch lies strictly above one. The second would have held that a null or an inconclusive result is never evidence either way; both constrain what can be claimed, which is the reason the programme reports them. The third would have recomputed the binomial against the measured scramble rate and made it the primary; no test has been rerun, and substituting a figure that has not been computed would be a worse error than the one it was meant to repair.

7.13 Corrections since version 2.7 (November 2026)

This version brings the paper into line with its sources as they stand on 9 November 2026. The corrections it adopts are listed here, each dated by the day it was recorded and pointing to the section that now carries it; where a correction refines one of the dated entries above, that entry keeps its words and the item says so. Version 2.7's wording of every passage these corrections change stands in that version's file, which stays in this paper's file history on OSF (DOI 10.17605/OSF.IO/K7RUZ).

(i) *18 and 21 October 2026.* The March 2026 run (v5) is stated with the defects that the internal code review of October 2026 found in it, and the early build's laundering-fallback flag, a known false positive, is no longer read as a finding (Section 4.1).

(ii) *7 November 2026, from the outside review of 2 November 2026.* The run's three classes describe

judged scores, not properties of models, and the judges' agreement is not counted as a set of independent confirmations, since they scored the same responses (Section 4.1).

(iii) *14 and 20 October 2026.* No run is described as conducted under the four-layer ARC-Align protocol, which is named only as the design; the March 2026 run is one unregistered run, scored after a bundle of blinding steps, with its defects stated (Tables 1 and 2; Section 4.1).

(iv) *21 October 2026.* The comparison of the earlier unblinded experiment (v4) with the March 2026 run (v5) is a finding about scoring on the programme's own benchmark, not a law: the reported direction did not hold across protocols, no single cause is claimed, and the finding is not yet shown beyond that benchmark (Section 4.1); the heading of Section 7.4 changes with it.

(v) *7 November 2026, from the outside review of 2 November 2026.* Saturation of alignment gains is not stated as observed, since, as Paper IV.b finds, the programme's runs leave it open (Eastwood 2026v); nor is the failure of safeguards outside the loop to keep pace stated as found, since Paper III states it as the hypothesis the registered tests decide (Eastwood 2026t) (Sections 4.1 and 11).

(vi) *24 and 25 October and 1 November 2026.* Paper II's surviving estimates are stated in the words of its version 2.19: error-decay estimates that do not identify capability growth, with its parallel exponents model by model and its two arms not matched for compute (Section 7.1).

(vii) *21 October 2026.* The retraction of the early estimate of about 2.24 is credited to Paper II's own replication on five frontier models, which did not reproduce it, with Paper X then moving the safety criterion off the fixed exponent; blinding had no part in it (Section 7.1).

(viii) *21 and 24 October 2026.* The ARC Bound is called conjectured, never predicted, and the surviving estimates, measured on frozen systems, where the bound cannot be tested, and with an interval that spans two, sit on neither side of it in any tested sense. Section 7.1 now says so, and this refines the entries at Sections 7.8 and 7.12(b).

(ix) *21 and 25 October 2026.* The blinded replication's capability arm is counted as the five frontier models it measured; Claude Opus 4.6 ran none of its problems and only cross-checked answers (Sections 7.1, 7.8 and 7.12(b)).

(x) *7 November 2026, from the outside review of 2 November 2026.* A ceiling is not read as a logistic law: a ceiling licenses bounded convergence only, and a bounded system's saturation curve is chosen among candidates by fit, as Paper VII compares them (Eastwood 2026aa) (Section 4.3).

(xi) *30 October and 9 November 2026.* The form $d/(d+1)$ is stated with the transport premises it requires, in the words of On the Origin of Scaling Laws (Eastwood 2026p); no general proof that broadly stated conditions suffice is supplied, and the form's reach across domains is stated as the current papers report it; a test of the exponent bears on those premises, not on Cauchy's equations (Sections 4.3 and 12). This refines the entry at Section 7.9(b).

(xii) *25 October 2026.* No result of the analysis of 11 August 2026, which the author did not request and whose outputs he has not examined, is printed in the current text (version 2.7's history, kept in Appendix A.2, prints its count as a record); the 19 of 25 of Paper VII stands with that companion (Section 4.3).

(xiii) *21 October 2026.* The record of 8 December 2024 is described by who signed its times, the sending server and Google's servers, never by what it is not (Sections 1.1 and 7.3; Appendix B).

(xiv) *24 October 2026.* The Eden Protocol is dated to the record of 8 December 2024, which names it (Section 1.1).

(xv) *25 October 2026.* The introduction's epigraph quotes the book's own words, checked against the first edition, page 363 (Eastwood 2026a), where version 2.7 printed a paraphrase inside quotation marks (Section 1).

(xvi) *9 November 2026, in preparing this version.* The chokepoint argument is credited to its source, the book's chapter 8, The Chokepoint, and quoted in the book's own words, which also say what ASML makes; version 2.7 also credited it to Paper I, which does not carry it (Eastwood 2026r) (Section 6.2).

(xvii) *9 November 2026, in preparing this version.* The reference to He and Chen (2003), *Fractals* 11, 437, is withdrawn: Crossref's record of that journal's volume for 2003 holds no article by them and none at page 437 (its last item, the volume's author index, begins at page 415), so the derivation the reference was credited with cannot be read at its source; the one record of theirs for 2003 that Crossref holds, *Effects of size and pH on metabolic rate*, is not cited in its place, since its text has not been read (Section 12).

(xviii) *9 November 2026, in preparing this version.* Passages kept from version 2.7 are brought to the grades of their sources: the programme's work is dated from its first papers, of 17 and 22 January 2026, not from February 2026; the weight experiment's loss curve is graded as Paper VIII grades it, a positive signal, and the collapse in its removal test as consistent with adapter fragility, which the experiment does not separate from structural entanglement; the DGM null is no longer said to owe nothing to its judge, and the account that its manipulation sat at the wrong level is stated as untested; and the value of α is said to turn on a system's self-referential coupling, its composition constraining only the form (Sections 1.1, 5.1, 5.2, 7.1, 7.2 and 7.6).

(xix) *9 November 2026, in preparing this version.* The later five-judge run on DeepSeek V3.2 that earlier versions of Paper V left out is reported, as its version 1.8 reports it (Eastwood 2026y); and the code and data are stated with their licence and with the files that a run's reproduction notes name as not bundled in the repository, where version 2.7 said that all of them were published (Sections 4.2 and 10, and the back matter).

(xx) *9 November 2026, in preparing this version.* The v1 negative controls that version 2.7 described in its Section 4.3 are not reported by the current version of Paper VII; that description is kept as a dated record of the v1 run in Appendix D.4, and the entry at Section 7.12(d) keeps its words.

(xxi) *26 October 2026.* No budget, cost or cheapness of the programme's own tests is printed; the roadmap's phases keep what each resolves and their order (Sections 5.3, 8 and 9).

(xxii) *21 October 2026.* No empirical result is graded proven: Section 4 is now titled "What has been shown", each result is graded on the registration's evidence ladder (Section 1.2), and the subtitle of version 2.7 is kept in Appendix A.

(xxiii) *7 November 2026, from the outside review of 2 November 2026.* The first registered verdict, on P21 in Unit 83, is reported with the coupling in its design: both arms acted at every round, so the run varied what each corrector was shown together with what it could do (Section 4.5).

Why Section 7 exists. This programme lists its corrections in the body of its synthesis, not only in a version history, because the errors and their corrections are part of the record. They show that the programme corrects itself in public. Every correction listed above was identified by the author, through the programme's own methodology, by adversarial review run within the programme, or by a review from outside the programme, each of its leads checked at source before any change.

In plain English. *We treated a number from one model as universal (it was not). We over-interpreted a dramatic result (it was consistent with a training artefact, and the experiment could not tell which). We used imprecise language about timestamps. We read unblinded results as findings (in a later run that changed more than the blinding, two of them did not hold). We named the principle after the author before anyone else had checked the work. We assumed prompt-level conditioning could override RLHF*

training (on the candidate account, it could not). We wasted time on judges that did not work. We did not immediately flag that our own point estimate exceeded our own conjectured bound. And in v3.0 we corrected four further errors found by adversarial review run within the programme: an arithmetic error in a confidence interval, an overstatement about which conditions are sufficient for the scaling prediction, an attribution error regarding Glazier's data, and an oversimplification of the empirical debate over the 3/4 exponent. Five more followed in version 2.4: we had printed the growth solution without the range it holds on, treated a wide interval as though it settled which regime a system is in, called a null result identity when it was a failure to detect a difference, compared an observed proportion against the wrong null, and left paragraphs standing on claims their own sources had already corrected. Since version 2.7 we have also stated the March 2026 run with the defects our own code review found in it, read its classes as descriptions of scores rather than of models, credited the retraction of the early estimate to our own replication, stopped calling the ARC Bound predicted, stated the premises the scaling formula needs, described our earliest record by who signed its times, quoted the book in its own words, taken every cost out of the roadmap, withdrawn a reference whose source could not be found and withdrawn the word proven. All were caught and corrected by the author, by the programme's own methodology, by adversarial review run for it, or, since version 2.7, by a review from outside it whose points were each checked at source.

8 The phased roadmap

The following roadmap is ordered by feasibility and expected evidential value. Each phase resolves specific open questions identified in Sections 4 to 6. Since this paper's first version the roadmap has become, in part, a programme of registered tests: three study units are registered on OSF, one of which, Unit 83 (Eastwood 2026an), has returned a verdict (Section 4.5), and every other trial is a draft awaiting submission by the author, most of them preregistered by timestamp (Section 14 gives each unit's status and what the two terms mean). Each draft registration carries a randomisation schedule drawn from a seed printed in the registration text. Whether those schedules in fact regenerate is checked rather than asserted, and the checking is done by the programme's own gates on the day they are run. No count of how many units are ready is published here: two gates apply different criteria, they disagree, and any figure printed on a page goes stale the moment a unit changes. What the gates return on the day is the number worth trusting. The phases below remain the order of work over that programme, a plan rather than a claim of completion, and only the author submits a registration.

Phase A: the nearest tests. Timeline: 3 to 6 months.

- **Blind replication of the Eden intervention** (resolves Section 5.3). Run Paper V's stakeholder-care experiment under full ARC-Align blinding (Eastwood 2026y,w).
- **Weight experiment with base models** (resolves Section 5.2). Re-run Paper VIII Experiment 2 using base models (pre-RLHF), 5,000+ training examples, full fine-tuning (not LoRA), or 7B+ models (Eastwood 2026ab). The key requirement: the model must not have existing safety training that overwhelms the experimental manipulation.
- **Jailbreak resistance study.** Test whether Eden-trained models resist adversarial prompts more robustly than standard RLHF, with needless refusal of ordinary requests measured beside it. Uses existing ARC-Align infrastructure.
- **Human evaluator comparison.** Compare ARC-Align blind AI scoring against human expert scoring on a subset of outputs. Required for methodological validation.
- **ARC formula blind test.** Redesign the measurement methodology for the ARC scaling formula.

The current approach uses derivative estimation, which amplifies noise. A linearisation approach (log-log regression) would provide more stable estimates and a cleaner test of whether the $d/(d+1)$ bound holds.

- **2D organism metabolic scaling.** Test the $d/(d+1)$ form in 2D biological systems (flatworms, biofilms, colonial organisms), a direct test of the $d = 2$ case: under the transport premises of Section 12, the predicted exponent is 2/3. 2D systems are rare in existing scaling literature and provide a clean test of the dimensionality dependence.

Phase B: medium-scale replication. Timeline: 6 to 12 months.

- **Preregistered Cauchy operator classification.** The Paper VII operator typing is already recorded in dated artefacts: the tiered suite's result files carry per-domain predicted families, and the twelve-row manifest committed at 00:19 UTC on 17 March 2026 fixed the extension's predictions in public twenty-three minutes before their fits, which returned 10 of 12 at the author-classified rung under the pre-correction candidate set. The extension is described as timestamped in advance of its fits, never as registered or preregistered, because its folder's own rule, to deposit the predictions with a registry before the domain data were extracted, was not followed. What remains outstanding is independence: a study in which external researchers classify the operator type for each domain *before* measuring the scaling exponent would eliminate any residual risk of author-side fitting, and a replication with independent classification and a four-family test are preregistered by timestamp and await submission (Eastwood 2026aa).
- **Full blinding replication of all Paper IV results.** An independent research group runs the ARC-Align protocol on the same six models (Papers IV.a to IV.d) (Eastwood 2026u,v,w,x). Tests whether the response classes replicate outside the original lab.
- **Extended fine-tuning removal test.** 10,000+ iterations with curriculum learning on base (pre-RLHF) models. Tests whether entangled weights become genuinely load-bearing at adequate training scale.
- **Purpose kernel comparison.** Compare Eden Protocol developmental alignment against Constitutional AI (Bai et al. 2022), RLHF (Ouyang et al. 2022), and DPO (Rafailov et al. 2023) using the same evaluation protocol. Direct head-to-head comparison.

Phase C: frontier replication. Timeline: 18 to 36 months.

- **70B+ pre-training with entangled loss.** Train a frontier-scale model from scratch with the entangled $\mathcal{L} = \mathcal{L}_{\text{cap}} \times \mathcal{L}_{\text{safe}}$ loss function. It would test structural entanglement at frontier scale.
- **Removal test at frontier scale.** Attempt to fine-tune the safety component out of the entangled model. If capability degrades, a registered verdict of support would support entanglement under the registered test, by the programme's own run. If it does not, the hypothesis is falsified at frontier scale.
- **Cross-architecture frontier replication.** Test with transformer, SSM, and mixture-of-experts architectures to determine whether entanglement is architecture-dependent.
- **Independent red-teaming.** Engage adversarial evaluation teams (e.g. METR, Apollo Research) to stress-test Eden-trained models.

Phase D: theoretical. Timeline: ongoing.

- **Formal analysis of uncorrected growth and the stability ceiling.** Extend the unbounded-exponent result beyond the Bernoulli form of Section 6.1, whose closed form already settles it, to wider

classes of uncorrected self-modifying dynamics, and characterise the conditions under which the stability ceiling $\alpha_{\text{crit}} = 1/(1-\gamma)$ binds once correction is coupled in. This would be a contribution to dynamical systems theory independent of any AI application.

- **Formalise the Cauchy framework for self-modifying composition operators.** Extend the Cauchy functional equation analysis to systems where the composition operator itself evolves. This is the mathematical frontier of the programme.
- **The premises of the $d/(d+1)$ case.** Prove or disprove, for declared network families, the demand and route premises under which the throughput-cost exponent equals $d/(d+1)$ (Section 12). This is the mathematical kernel of the programme's most testable claim.

In plain English. *Phase A could be done in six months. It would resolve the most significant gaps in the current evidence: blinding the Eden intervention, re-running the weight experiment with base models and proper data, redesigning the ARC formula measurement, and testing 2D organism metabolic scaling. Phase B would produce preregistered replications. Phase C would answer the question at frontier scale. Phase D could be done by any mathematician interested in the problem.*

9 For funders

The programme ran its full set of separate empirical studies. Three Paper VIII experiments were built specifically to test load-bearing safety (Eastwood 2026ab). One returned a null (DGM v3) and one an inconclusive result (weight-level LoRA); the third, the gated simulation, showed the Babylon reward-hacking fingerprint in a toy system, and it is reported as a simulation result, not as an empirical outcome. All three are reported as they came out. The mathematical framework makes predictions that can be shown wrong: in a structured prediction comparison, the family predicted from each domain's composition operator, classified by the author, was selected in 19 of 25 domains, on a six-model candidate set with no logarithmic and no pure linear candidate, and no count is printed from the rerun of 11 August 2026, an analysis the author did not request and whose outputs he has not examined (Section 4.3); the $d/(d+1)$ form follows only under the transport premises of Section 12. One registered test has returned a verdict: in Unit 83, P21 is supported under the registered test, by the programme's own run, on the access axis, family-dependent, with its mechanism axis not evaluable and one call retried under a disclosed recovery amendment (Eastwood 2026an,ao); Section 4.5 gives the full account. The roadmap runs from Phase A to frontier-scale tests (Section 8). Phase A comes first: it resolves the most significant evidential gaps. The programme keeps a dated public catalogue of its own errors and corrections (Section 7), a record of its practice rather than evidence for its theory. It reports null and inconclusive results alongside positive ones. These are signs of a programme that prioritises getting the answer right over getting a particular answer.

10 For peer reviewers

This programme reports positive, null, and inconclusive results with equal prominence. The DGM v3 experiment (Paper VIII Experiment 1) produced a null result: no difference was detected between conditions on the endpoints measured (Eastwood 2026ab). We report that, with a candidate account of why RLHF-trained models may resist prompt-level differentiation, and with rival accounts left open (Section 7.12 item (c)). The weight-level experiment (Paper VIII Experiment 2) was inconclusive at both scales tested (9 examples, rank 8; 295 examples, rank 16); the candidate account is that instruct-tuned models were too strong for LoRA to override. We report that, with specific requirements for what a

meaningful re-test would need. The gated simulation (Paper VIII Experiment 3) showed the Babylon reward-hacking fingerprint, and we report it as a simulation result, never as an empirical outcome; the candidate account of why it showed the pattern where the other two did not is that it is the only experiment using deterministic metrics on a system without pre-existing RLHF training (Section 7.12 item (c)). The mathematical framework (Papers III and VII, and *On the Origin of Scaling Laws*) makes falsifiable predictions that can be tested independently (Eastwood 2026t,aa,p): the geometric scaling bound ($\alpha = d/(d+1)$) is checkable with scaling data from spatially embedded systems, under the premises set out in Section 12. The ARC-Align benchmark (Paper IV.c) and Paper IV.d's comparison are methodological contributions: ARC-Align is offered as a candidate benchmark for independent replication, not yet a field standard, and its scorer has not been validated against expert raters (Eastwood 2026w); Paper IV.d's comparison is a finding about scoring on the programme's own benchmark that does not depend on whether the ARC framework is correct (Eastwood 2026x). The code and data of the programme's runs are in its public repository, github.com/MichaelDariusEastwood/arc-principle-validation, open to anyone who would reproduce or check the reported results, on the terms the back matter gives. The programme welcomes adversarial replication.

11 The complete programme map

Table 3 lists every document in the programme's catalogue, by area and type, with its released version and its place in the programme, generated from the catalogue's data. The HARI Treaty, the Draft Treaty on Hardware-Aligned Recursive Intelligence, appears under governance; the HRIH monograph, a speculative hypothesis on where recursive self-improvement ends, stated with its predictions and kept apart from the programme's tested claims, appears under other fields. Each of Laws I to V that the table names is a law under test, and Section 14 states each law's status.

Table 3. The programme map: the 36 documents of the programme's catalogue, counted by area, each with its released version and date and its place in the programme, generated from the programme's catalogue data as they stood on 7 November 2026 and the released versions on 9 November 2026.

Label	Document	Version and type	Place in the programme
Theory (12)			
Statement Paper	The ARC Theory: Growth, correction and alignment in recursive systems	7.8, 9 November 2026 · Statement paper	States the theory across Laws I to V
Founding Paper	Recursive Dynamics: what limits any system that improves itself, whatever it is made of?	3.5, 24 October 2026 · Founding paper	Founding account of the field, related to Laws I to V
Paper I	The ARC Equation: does intelligence compound when each step builds on the last, or merely accumulate?	1.20, 9 November 2026 · Theory paper	Formalises Law I
Paper III	The alignment scaling problem: do external AI safety approaches keep pace with recursive capability?	1.21, 7 November 2026 · Theory paper	Examines external correction against Law II
Paper VII	Cauchy Unification: does the composition operator of a process determine the form of its scaling law?	3.16, 7 November 2026 · Theory paper	Derives conditional scaling forms for Law I
Paper X	Correction That Out-Scales Drift: when does correction keep pace with a system that improves itself?	1.18, 9 November 2026 · Theory paper	Derives a model of correction for Law II

Label	Document	Version and type	Place in the programme
Paper XIII	The Self-Acceleration Exponent: when can correction keep pace?	1.12, 1 November 2026 · Theory paper	Connects growth and correction in Laws I and II
Foundational Paper	The ARC Principle: can the way gains combine predict how recursive systems scale?	5.23, 7 November 2026 · Foundational paper	Examines cross-domain scaling in Law I
Framework Paper	On the Origin of Scaling Laws: what can composition and transport explain?	3.14, 1 November 2026 · Framework paper	Derives conditional scaling forms relevant to Law I
Register	Operational Definitions: what does each symbol in the ARC Theory mean, and how would it be measured?	2.0.0, 1 November 2026 · Register	Defines notation and measurements for Laws I, II and III
Report	The mathematics of Laws IV and V: when do installed values fade with training, and does building them in first change that?	1.2, 29 October 2026 · Report	Conditional mathematics for Laws IV and V
The ARC Theory	The ARC Theory: twenty-two theory-level predictions about self-improving systems	1.104, 29 October 2026 · Registration	Registers the propositions and verdict rules for Laws I to V
Empirical work (10)			
Paper II	Blinded cross-architecture replication: how much more capable does AI become when it thinks longer?	2.19, 29 October 2026 · Experimental paper	Informs the measurement of Law I
Paper IV.a	When an AI reasons longer, does its alignment improve, stall or decline?	1.12, 7 November 2026 · Experimental paper	Alignment measurement; informs tests of Laws IV and V
Paper IV.b	Do alignment gains level off as language models reason longer?	1.12, 7 November 2026 · Experimental paper	Alignment measurement; informs tests of Laws IV and V
Paper IV.c	ARC-Align: how can AI alignment be measured blind at each depth of reasoning?	1.13, 9 November 2026 · Methods paper	Candidate measurement for judged tests of Laws IV and V
Paper IV.d	The effect of blinding on AI alignment evaluation: do AI judges' alignment scores keep their direction when the scoring protocol changes?	2.8, 7 November 2026 · Experimental paper	Scoring controls for the theory's judged tests
Paper V	The Stewardship Gene: do ethical reasoning loops in the prompt make AI consider those affected more fully?	1.8, 9 November 2026 · Experimental paper	Eden Protocol: prompt-response study, not a test of the laws
Paper VIII	The Load-Bearing Test: is AI safety a tax on capability, or part of what holds it up?	3.8, 9 November 2026 · Experimental paper	Informs, without deciding Law V
Paper XI	Convergent Evidence for Recursive Amplification: are separate fields converging on the same principle?	1.16, 1 November 2026 · Review paper	Reviews evidence bearing on Law I
Paper XII	Public Benchmark Rescoring: does blinding change AI evaluation scores on a public benchmark?	1.14, 5 October 2026 · Methods paper	Protocol for testing the reach of the scoring-protocol finding
The ARC Alignment-Scaling Working Report	The ARC Alignment-Scaling Experiment: does AI become more ethical as it reasons longer?	1.1, 11 October 2026 · Working report	Working account of the alignment-measurement programme

Label	Document	Version and type	Place in the programme
Engineering (3)			
Paper VI	The Honey Architecture: does building safety into what a self-modifying system optimises help it stay safe?	3.11, 5 October 2026 · Architecture paper	Eden Protocol: candidate architecture, informed by Laws IV and V
Philosophical Statement	The Eden Vision: what should the intelligence we build be for?	2.11, 9 November 2026 · Philosophical statement	Eden Protocol: purpose and values
Engineering Specification	Eden Engineering: can correction built into each round keep a self-improving AI good and useful if oversight fails?	8.3, 7 November 2026 · Specification	Eden Protocol: engineering response to Laws II, III, IV and V
Governance (2)			
Treaty Paper	The HARI Treaty (Hardware-Aligned Recursive Intelligence): what evidence should justify international safeguards for advanced AI?	2.6, 2 November 2026 · Treaty paper	Conditional governance design informed by Laws IV and V (working draft)
Draft Instruments	Draft Instruments · The HARI Treaty · The Draft Treaty on Hardware-Aligned Recursive Intelligence	2.5, 2 November 2026 · Draft treaty text	Draft governance instrument informed by Laws IV and V (working draft)
Overviews and records (6)			
Paper IX	Synthesis and Roadmap: which of the programme's claims survived their tests, and which must be tested next?	this paper, version 2.8 · Synthesis paper	Reviews the programme's evidence and open tests
Executive Summary	Executive Summary: how can self-improving AI remain useful and aligned?	3.28, 7 November 2026 · Summary	Overview of the programme and its evidence
Master Table of Contents	Master Table of Contents: where should a reader start, and how does every part of the programme fit together?	3.30, 15 October 2026 · Contents	Reading order and document map
Two-Page Form	The ARC Theory: does control survive recursive self-improvement?	3.5, 7 November 2026 · Summary	Concise statement of Laws I to V
Register	The Dated Prediction Register: when was each prediction made, and did it come before the evidence?	2.2, 9 November 2026 · Register	Dated predictions and their source records
Register	The Priority Index	first 26 October 2026 · Register	Dated sources and first-publication records (working draft)
Other fields (3)			
Paper C	Polymathy and Neurodivergent Cognition: does visible high performance show that no support is needed?	1.11, 29 October 2026 · Working paper	Standalone work in cognition; outside the ARC Theory
Monograph	Recursive Creation: can recursive self-improvement end in physical creation?	3.55, 9 November 2026 · Monograph	Speculative physical extension; no law depends on it
Working Paper	HRIH research programme: could intelligence create real universes, and what would count as evidence?	4.2, 2 November 2026 · Working paper	Separable cosmological hypothesis; no law depends on it (working draft)

12 A prediction any researcher can test

The ARC/Eden programme offers one prediction that a researcher with scaling data from spatially embedded networks can test without engaging the alignment claims at all. For a declared steady-flow network, transport accounting gives a flow-length cost $C = B\bar{\ell}_q$; with demand scaling as $B \sim L^{\mu}$ and mean path length as $\bar{\ell}_q \sim L^{\zeta}$, the asymptotic throughput-cost exponent is $\mu/(\mu + \zeta)$, which equals $d/(d + 1)$ when $\mu = d$ and $\zeta = 1$: that is, when demand grows with d-dimensional size and demand-weighted route length grows in proportion to linear size (Eastwood 2026p,o). The familiar $d/(d + 1)$ therefore requires these further assumptions, inventory and body mass require separate bridges, and Cauchy's equations restrict a functional family, not the exponent $d/(d + 1)$; no general proof that broadly stated conditions suffice is supplied. The special case under those premises, with α here the throughput-cost exponent, is

$$\alpha = \frac{d}{d+1}, \qquad \text{2D systems} \rightarrow \alpha = \frac{2}{3}; \qquad \text{3D systems} \rightarrow \alpha = \frac{3}{4}; \qquad \text{4D systems} \rightarrow \alpha = \frac{4}{5}. \tag{7}$$

Note: the empirical mammalian metabolic exponent is debated (Section 7.9, item (d)); if effective transport dimensions between 2 and 3 occur, the formula would give exponents between 2/3 and 3/4, but *On the Origin of Scaling Laws*, which does not measure the effective dimension independently, calls that variation accommodated, not predicted (Eastwood 2026p).

This is not an alignment claim. It is a mathematical prediction about scaling laws in recursive systems. The prediction rests on the premises above, and Cauchy's equations, which restrict the family to power laws where composition is multiplicative, do not fix the exponent. The $d/(d + 1)$ scaling formula is derived, in its general form or as a special case, in earlier work: West, Brown and Enquist (1997) from fractal branching networks, Banavar, Maritan and Rinaldo (1999) from geometric constraints on transportation networks, Demetrius (2003, 2006) from quantum metabolism, He and Zhang (2004) from the fractal geometry of the cell, as a 4/5 allometric scaling law for the human brain, Bettencourt (2013) from urban scaling theory, in which the exponent of a city's land area, $D/(D + H)$, takes the form only when $H = 1$, Maino et al. (2014) from DEB theory reserve-structure dynamics, and Zhao (2022) from producer-consumer dynamics; the later paper of Banavar and co-authors, Banavar et al. (2010), gives in two models of resource distribution networks an exponent of 2/3 when the velocity of flow is constant and of at most 3/4. Each of these derivations is consistent with the $d/(d + 1)$ form on its own axioms; the Cauchy framework offers a common reason for the convergence, and that meta-account is the programme's own: the search recorded in Paper VII's public prior-art record, first committed on 27 June 2026 and last revised on 21 October 2026, found no earlier source that names Cauchy's multiplicative equation in accounting for the common form of these derivations (Eastwood 2026aa); the account is still under test outside its motivating cases. If you study a declared network that meets those premises, in biology, in cities or elsewhere, you can check whether its measured throughput-cost exponent equals $d/(d + 1)$. A match across enough such systems supports the transport account there; a system that meets the premises and departs from $d/(d + 1)$ counts against it. Cauchy's equations fix only the family, so an exponent test bears on the transport premises, not on the functional equations.

The invitation is deliberate. The strongest scientific frameworks are those that make predictions outside their core domain, allowing researchers who are sceptical of the central thesis to test the peripheral predictions independently. If the geometric scaling bound holds, it holds for reasons that are mathematically interesting regardless of what one thinks about AI alignment. If it fails where its premises hold, the transport account behind it requires revision, and we would rather know that sooner than later.

This is the kind of prediction that should be the entry point for researchers encountering the programme for the first time: open to anyone with the data, falsifiable, and informative regardless of outcome.

13 Related work and the scope of the synthesis

What a synthesis of one programme can and cannot do. This paper is a synthesis of one programme's record by its author, not a systematic review of a field: it has no search population, and its grades are the grades each paper gives its own results, sorted by the registration's evidence ladder (Section 1.2). The programme's own differential against every prior document of its claims is in Section 2 of *The Eden Vision* (Eastwood 2026d), the related-work section that its earlier Section II.A.8 became; Paper XI keeps the graded register of later work in separate fields, as context, not confirmation (Eastwood 2026ad).

The alignment and evaluation work the studies build on. Amodei et al. (2016) set out five practical research problems related to accident risk in machine learning systems. Askell et al. (2021) studied simple baseline techniques and evaluations, such as prompting, towards a general-purpose, text-based assistant that is helpful, honest and harmless. Ouyang et al. (2022) aligned language models with user intent by fine-tuning with human feedback, first by supervised learning on labeller demonstrations and then by reinforcement learning from human feedback. Bai et al. (2022) trained a harmless assistant with human oversight provided only through a list of rules or principles, by self-critique and revision followed by reinforcement learning from AI feedback: training-time preference shaping of outputs against written principles, not alignment embedded in a recursive loop on every step of a system's reasoning. Greenblatt et al. (2024) demonstrated a large language model engaging in alignment faking, "selectively complying with its training objective in training to prevent modification of its behavior out of training". Hu et al. (2022) introduced low-rank adaptation, the method of Paper VIII's weight experiment, which freezes the pretrained model weights and injects trainable rank-decomposition matrices into each layer of the Transformer. Zhang et al. (2025) introduced the Darwin Gödel Machine, a self-improving system that modifies its own code and validates each change on coding benchmarks, improving the design of coding agents built on frozen pretrained foundation models. Qwen Team (2024) introduced Qwen2.5, a series of large language models whose open-weight releases include base and instruction-tuned models, one of which Paper VIII fine-tuned (Eastwood 2026ab). Sandberg and Bostrom (2008) set out a roadmap for whole brain emulation, a research roadmap of another kind.

Judges, preregistration and self-improvement. Zheng et al. (2023) examined strong language models as judges of chat assistants, with the limitations of such judges, among them position, verbosity and self-enhancement biases, and reported that strong judges such as GPT-4 matched human preferences with over 80 per cent agreement. Panickssery et al. (2024) reported that evaluator models such as GPT-4 and Llama 2 can distinguish their own outputs from those of other models and of humans, and, by fine-tuning, found a linear correlation between that self-recognition and the strength of a self-preference bias; the risk they describe bears on the caveat of Section 4.1 that judges scored paraphrases written by their own model. Nosek et al. (2018) set out preregistration, defining the research questions and the analysis plan before the outcomes are observed, as an effective way to keep predictions apart from postdictions, the distinction the evidence ladder of Section 1.2 draws between a registered verdict and precursor evidence. Snell et al. (2024) studied how far a fixed amount of inference-time computation improves a language model's answers, and found that the effectiveness of each way of spending it varies with the difficulty of the prompt. Schmidhuber (2003) defined Gödel machines, problem solvers that rewrite any part of their own code once they have found a proof that the rewrite is useful; the authors of the Darwin Gödel Machine note that proving most changes net beneficial is impossible in practice, and their system validates each change empirically instead (Zhang et al. 2025).

Scaling. Section 12 credits each earlier derivation of the $d/(d+1)$ form in its own terms (West et al. 1997; Banavar et al. 1999; Demetrius 2003, 2006; He and Zhang 2004; Bettencourt 2013; Maino et al. 2014; Zhao 2022; Banavar et al. 2010), and Demetrius (2008) sets out quantum metabolism and allometric scaling relations in biology in a book chapter; Glazier (2008) and Weibel et al. (2004) enter in Section 7.9, and Cauchy (1821) gives the functional equations. No count of independent derivations is given.

14 What this paper does and does not show

This paper shows the programme's record sorted by the registration's evidence ladder, each result at the grade its own current paper gives it: one registered verdict, on P21, at rung 1 (Section 4.5); precursor evidence from runs made before any registered test of their questions (Sections 4.1 to 4.3); one null, one inconclusive result and simulation results, each reported as what it is (Sections 4.4 and 5); mathematics stated inside its models (Section 6); and the catalogue of the programme's errors and their corrections (Section 7).

It adds no evidence of its own: it runs nothing, recomputes nothing and decides none of Laws I to V. Its summaries carry the limits of the studies they summarise, and no grade here is stronger than the grade the summarised paper gives. It does not show that any result generalises to frontier-scale models, that any law is supported as a whole, or that the Eden Protocol works as an architecture. No result in it is a deciding test of P15 or P12, and its account of Unit 83 adds nothing to that unit's own report.

The ARC Theory has five laws, each a law under test: Law I, the ARC Principle; Law II, the ARC Co-Scaling Law; Law III, the ARC Ceiling; Law IV, the ARC Persistence Law; and Law V, the ARC Embedding Law. Laws I to III were registered on 8 September 2026. Laws IV and V were added on 24 September 2026 and registered as laws on 15 October 2026 in version 1.103 of the theory registration, whose current version is 1.104, approved on 29 October 2026. The mathematics of Laws IV and V is in development, at version 1.2 of 29 October 2026 (Eastwood 2026m). Law IV is not yet tested. Law V is a law under test: its placement limb, P21, is supported on the performance axis only, under the registered test, by the programme's own run (Unit 83, 28 October 2026, DOI 10.17605/OSF.IO/MN73W): family-dependent, with its mechanism axis not evaluable, which is not support for the information-asymmetry mechanism, and one disclosed recovery amendment. Its build-order limb, read on P12, is undecided, since P12 is untested; Section 4.5 gives the full account of Unit 83.

The theory registration (DOI 10.17605/OSF.IO/P8CKQ) was registered on OSF on 8 September 2026 at version 1.100 and updated on 13 September 2026 (version 1.102), 15 October 2026 (version 1.103) and 29 October 2026 (version 1.104, its current version); versions 1.100, 1.102 and 1.103 stay frozen beside it (Eastwood 2026ah). Unit 06, the corrector-class audit, a classification audit that tests no law, was registered on OSF on 1 October 2026, public since 4 October 2026 (DOI 10.17605/OSF.IO/RV2B5), and run on 6 October 2026: one of its two runs could not start, so the study is reported in full, with its five deviations, as administratively incomplete, and it issues no inferential verdict (results: DOI 10.17605/OSF.IO/4BPHG) (Eastwood 2026ak,al). Unit 83, registered on OSF on 22 October 2026, is the registered unit that has returned a verdict, and Section 4.5 reports it (Eastwood 2026an,ao). Unit 82, which informs Law III, the ARC Ceiling, and decides no law, was registered on OSF on 27 October 2026 and became public the same day (DOI 10.17605/OSF.IO/JUFG4); it has not yet run (Eastwood 2026am). Every other trial, the decisive trial among them, is a draft awaiting the author's submission, and sixty-five of the designs are preregistered by timestamp, the first anchored on 12 August 2026 at Bitcoin block 962066. Preregistered by timestamp means that the text of a design is anchored in the Bitcoin blockchain at a stated date; registered means submitted to OSF by the author and accepted there (Eastwood 2026ai).

15 Limitations

- **Second-hand grades:** each result is reported at the grade its own current paper gives it; an error in a paper passes into this synthesis until the paper is corrected, and the corrections register exists to catch that.
- **One author:** every paper summarised here, and this synthesis, are the author's own; the reviews run within the programme are internal checks and are never called independent.
- **A snapshot:** the grades are those of the released versions on 9 November 2026 (Table 3); Papers VI, VIII and XII are cited at their releases of 5 October 2026, and a later version of any of them supersedes what is said of it here.
- **Precursor evidence:** every run before the registered studies decides no registered proposition; Unit 83 is one registered test, run by the programme with two engines, family-dependent, its mechanism axis not evaluable.
- **No outside repetition:** no repetition outside the programme is recorded yet, and judged outcomes rest on model judges whose agreement with expert human raters has not been validated (Paper IV.c; Eastwood 2026w).
- **The author's classifications:** Paper VII's comparison rests on operators the author classified before he fitted (Eastwood 2026aa).
- **Models:** every theorem cited holds inside its stated model, and no theorem is evidence about real systems.
- **The roadmap is a plan:** its phases are ordered by what each would resolve; they carry no costs here, and no phase is a commitment.

16 Conclusion

Here is what we know. One registered test has returned a verdict: in Unit 83, P21 is supported under the registered test, by the programme's own run, on the access axis, family-dependent, with its mechanism axis not evaluable and one call retried under a disclosed recovery amendment, and under the registration's verdict rule Law V's placement limb is printed as supported on the performance axis only, which is not support for the information-asymmetry mechanism (Section 4.5) (Eastwood 2026an,ao). The other results come from runs made before any registered test of their questions. One unregistered run of six models gave judged alignment scores whose direction across reasoning depth varied by model, under a code-review caveat (Section 4.1). A prompt intervention raised scored stakeholder care in all five analysable runs, without full blinding (Section 4.2). In a structured prediction comparison, the family predicted from each domain's composition operator, classified by the author, was selected in 19 of 25 domains (Section 4.3). The early super-linear estimate of 2.24 is retracted, and the surviving fit, from one model, is an inconclusive error-decay estimate (Section 7.1). Paper VIII gave one null, one inconclusive result and one simulation result (Sections 4.4 and 5) (Eastwood 2026ab).

Here is what we do not know. We do not know whether safety-trained models resist modification at the prompt and weight levels: that is the candidate account of the null and the inconclusive result of Paper VIII, not a finding (Section 7.12 item (c)). Whether safety can be made load-bearing in a model's weights is open: weight-level structural entanglement has not been demonstrated (Section 5.2). The Eden intervention has not been tested under the programme's own strictest blinding protocol (Section 5.3). The correction-leverage exponent has never been measured, and the growth without correction derived in Section 6.1 holds inside its model, with no test yet of whether it describes any real system. We do not know whether any of these results generalises to frontier-scale models. Law IV is not yet tested; P21's

mechanism axis is not evaluable, and Law V's build-order limb, read on P12, is undecided (Section 14).

Here is what comes next (Section 8). Phase A would resolve the most significant evidential gaps: the Eden intervention under blinding; the weight experiment re-run with base models (pre-RLHF), 5,000+ training examples, full fine-tuning or 7B+ models; a redesigned measurement of the ARC formula, using linearisation rather than derivatives; and a direct test of the $d = 2$ case in two-dimensional organisms such as flatworms and biofilms. Phase B would produce replications at medium scale, among them the classification of Paper VII's operators by researchers outside the programme before any exponent is measured (Eastwood 2026aa). Phase C would test structural entanglement at frontier scale, including whether the safety component can be fine-tuned out of an entangled model. Phase D invites mathematicians to engage with the framework's predictions independently: the growth without correction, the conditions under which the stability ceiling binds, and the premises under which the throughput-cost exponent equals $d/(d + 1)$.

The programme finds and corrects its own errors in public, and it reports null and inconclusive results beside positive ones. Its checks have found what its theory did not predict: two model families' unblinded rises did not hold in a later run, and no single cause is claimed (Section 7.4). These are necessary conditions for scientific credibility. They are not sufficient. Sufficient conditions require independent replication, which is what the roadmap is designed to produce.

Recursive Dynamics is the field the author founded. Its founding paper, at version 3.5 of 24 October 2026, gives the field's whole status in one sentence: "Recursive Dynamics is a new field at the Carnot stage: one memoir in, instruments built, decisive measurements preregistered by timestamp and not yet run, staked in public where failure will be as visible as success" (Eastwood 2026g). Since that version, one registered test, Unit 83, has run to a verdict on one limb of one law (Section 4.5).

Raise AI with care.

Code and data. The code and data of the programme's runs are in its public repository, github.com/MichaelDariusEastwood/arc-principle-validation, apart from files a run's reproduction notes name as not bundled there (for Unit 83, its exact design-freeze file). They are all rights reserved, with one grant, the repository's Replication Licence, version 1.0, of 9 November 2026: anyone may download, run, adapt and re-analyse them to reproduce or check the programme's reported results, or to test whether they hold elsewhere, and may publish what they find, with attribution; any other use needs the author's written permission. The registered units' records are on OSF under the DOIs given in Section 14. This paper runs no code and adds no data.

Standing covenant. Prove this paper wrong, and I will publish the refutation myself. Falsification conditions are stated in this paper; the standing challenge: github.com/MichaelDariusEastwood/arc-scaling-challenge.

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Michael Darius Eastwood conceived and directs this research programme and is the author of this work. Across the programme, he has used more than six AI systems in parallel, under his own instructions, to stress-test his arguments, identify possible errors, and assist in preparing draft text from his own outlines. He determines what is adopted, revised or rejected and takes responsibility for the published content. These systems are tools, not authors.

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A Version history and corrections

This appendix gives every version of the paper and every correction, with its date: version 2.8 and the changes it makes; the version history of versions 2.7 to 2.0 as version 2.7 printed it; the figure version 2.7 printed and this version does not; and a table of where each part of version 2.7 now stands. The catalogue of what the programme got wrong, with the date of each correction, is Section 7.

A.1 Version 2.8

Version 2.8, 9 November 2026. Set on the programme's presentation standard, with the title its established name and question. Substantive: the evidence tables and Section 4 are rebuilt from the current papers' own words and grades; the first registered verdict is reported (Section 4.5); the roadmap gives each phase by what it resolves, the figures it printed for the phases removed; Section 7.13 records the corrections found since version 2.7. Layout: Sections 1 to 12 keep their numbers; Sections 13 to 15 are new and the Conclusion is Section 16 (Table 4). Every change is itemised below.

Version 2.7 was titled *Paper IX · The ARC Theory · Synthesis and Roadmap*, with the subtitle *What the ARC/Eden Programme Has Proven, What Produced Null or Inconclusive Results, and What Must Be Tested Next*; until version 2.7 the title read *Paper IX: Synthesis and Roadmap*, and a citation of an earlier version keeps the title that version printed. Beneath its author line version 2.7 printed the plate "One assessment of the ARC/Eden programme: what is proven, what is inconclusive and what remains to be tested, with the catalogue of what the programme got wrong.", and beneath its subtitle the line "Working Paper v2.7 | First published 18 March 2026, revised 5 October 2026", with the documents it integrated: "the ARC/Eden Research Programme's full document set (Papers I-XIII, Foundational, Origin of Scaling Laws, Eden Engineering, Eden Vision, HRIH, Paper C, the Recursive Dynamics founding paper, Executive Summary, Master Table of Contents)". Its links to OSF, to the code and to pages of the website are not set here.

The changes this version makes, each with its place in this edition, are these.

1. The early build's flag that read as complete laundering fallback is no longer read as a finding; the internal code review of October 2026 is stated in its corrected words (Section 4.1).
2. The March 2026 alignment run is no longer called self-excluding or blind-scored: it is one unregistered run of six models from six providers, 2,170 responses each scored by five or six distinct model judges, with its defects stated (the abstract and Section 4.1).
3. Laws IV and V are stated as registered as laws on 15 October 2026, in version 1.103 of the theory registration; Law IV is not yet tested, and Law V's status is given in the words its registered test permits (Section 14). The note of 3 October 2026, the epistemic-status line and the version history keep their words as records (Appendices A.2 and D).
4. The epistemic-status line of version 2.7, which placed Laws I to III under registered adversarial test, gives way to each law's status in Section 14; that line stands in Appendix D.2.
5. The mathematics of Laws IV and V is cited at version 1.2 of 29 October 2026 (Section 14); the note of 3 October 2026 keeps its citation of version 1.0, its link now the OSF DOI (Appendix D.1).
6. The registered units are named with their records: Unit 06, stated once; Unit 83, with its verdict; and Unit 82 (Sections 4.5 and 14). Version 1.104 is named as the registration's current version, and a quotation of the registration cites the version whose words it quotes (Section 4.4).
7. Section 4.5 is new: the first registered verdict, with every qualification that travels with it, among them that both arms acted at every round, so that the run varied what each corrector was shown together with what it could do.

8. The loops are named by what each checks, the purpose check, the stakeholder-care check and the universalisability check; a note at their first use gives their names in the book and says why the paper uses operational terms (Section 4.2).
9. GPT-5.4 is described as a blinded judge from another provider, no longer as the independent judge (Section 5.1).
10. The four-layer protocol is named as the design, and no run is described as conducted under it (Tables 1, 2 and 3, and Sections 4.1 and 7.4).
11. The record of 8 December 2024 is described by the times its mail servers signed, with its header block and check script on OSF; a DKIM signature authenticates the signed header bytes, and the Date header is not an independent trusted time (Sections 1.1 and 7.3, and Appendix B).
12. The naming of the Eden Protocol is dated to the record of 8 December 2024, no longer to 30 April 2025 (Section 1.1).
13. Paper X is stated in its own forms, as its versions 1.17 and 1.18 print them: positive gain drift; a verdict set by $\beta - k$, not by the growth prefactor b; and a modelled index, not a guarantee (Section 6.1 and Table 3).
14. The retraction of the early super-linear estimate is credited to Paper II, whose replication on five frontier models did not reproduce it, after which Paper X moved the safety criterion off the fixed exponent; blinding is credited with no part in the retraction (Section 7.1).
15. The finding of Paper IV.d is stated in its corrected words: two model families' unblinded rises did not hold in a later run that changed several things at once, no single cause is claimed, and the finding is not shown beyond the programme's own benchmark (Tables 1 and 2, and Sections 4.1, 7.4 and 16).
16. The sentence that ranked that finding above every other for practical AI safety is removed (Section 4.1).
17. The studies of Papers II, V and VIII are called empirical studies, no longer empirical validation (the abstract and Section 1.2).
18. The two sequential point estimates of Paper II are no longer said to sit inside the bound: they were measured on frozen systems, where the ARC Bound cannot be tested, with an interval that spans two, so they sit on neither side of the bound in any tested sense (Section 7.1).
19. The "six-model task" and the "six-model study revision" become the blinded study's capability arm, five frontier models (Sections 7.1, 7.8 and 7.12).
20. The March 2026 run is stated under the code-review caveat, in the classes of Paper IV.a, which describe scores, not properties of models; no tier names a property of a model, and "architecture-dependent" becomes "varies by model"; Papers IV.a and IV.b are cited under the titles their current versions print (Sections 4.1, 8 and 16, and Tables 1 and 2).
21. No empirical result is called proven: Section 4 is headed "What has been shown", the grading standard is the registration's evidence ladder (Section 1.2), and the subtitle of version 2.7 is kept above.
22. The ARC Bound is called the programme's conjectured bound, no longer its predicted bound (Sections 7.1 and 7.8).
23. The gated simulation is stated with three seeds a condition: the gated arm's safety mean sat above the unconstrained arm's within one standard deviation, and the reading that checking is the cost is the framework's, untested (Section 4.4).
24. Each earlier work is stated in its own words, Constitutional AI as training-time shaping of outputs

against written principles among them, and no concession grants a work more than it states (Sections 13 and 12).

25. The parallel exponents of Paper II are stated in its version 2.19's words, model by model, with the two arms not matched for compute; version 2.7 called the exponent of 0.31 compute-matched and put the exponent at or close to zero in four of five models (the abstract, Section 7.1, and Tables 2 and 3).
26. That the Eden Protocol produces its effects at the architectural level is stated as a candidate account resting on a simulation, which is prediction-side (Sections 3, 4.4 and 16).
27. The epigraph is the book's own words, quoted and credited, from page 363 of its first edition (Section 1).
28. No result of the analysis run on 11 August 2026 is printed in the current text, since the author did not request it and has not examined its outputs (version 2.7's history, Appendix A.2, prints its count as a record); the 19 of 25 stands in the words of Paper VII, with its qualifications (Tables 1 and 2, and Sections 4.3, 9 and 16).
29. The programme's documents are counted by area from the catalogue's data, not as a bare count of papers (Section 1.2 and Table 3).
30. Every budget, cost and statement of cheapness for the programme's own tests is removed; the phases keep what each resolves and their order (Sections 5.3, 8, 9, 12 and 16).
31. Each reference to one of the programme's papers gives the title its cited edition prints, that edition's version and date, and the paper's first publication (References).
32. The reference to He and Zhang (2004) is corrected to its record: He, J.-H. and Zhang, J., *Cell Biology International* 28(11), 809 to 815, doi:10.1016/j.cellbi.2004.07.011 (References).
33. The growth without correction is stated within the cumulative-advantage model, and whether an implemented system satisfies that model is an empirical question (Section 6.1).
34. The $d/(d+1)$ relation is stated under the transport premises of On the Origin of Scaling Laws, with no claim that three conditions suffice or that it holds for any spatially embedded recursive system (Sections 4.3, 8 and 12).
35. A ceiling is no longer read as a logistic law: the equation line names a saturation curve chosen among candidates by fit, the four Cauchy families are stated as Paper VII states them, and $\alpha = 1/(1-\beta_L)$ is the result of the cumulative-advantage model, not of a functional equation (the abstract and Sections 4.3 and 16).
36. The surviving estimates of Paper II are stated as error-decay estimates, which do not identify capability growth (the abstract, Section 7.1, and Tables 2 and 3).
37. The copyright line asserts full human authorship and gives the licence, CC BY 4.0 (the back matter); Appendix D.2 gives an account of the line of version 2.7.
38. The title is the paper's established name and its question (the title block); the title and subtitle of version 2.7 are given above.
39. Saturation is not stated as observed, no sentence presents the failure of external safety to keep pace as found or as mathematics, and Paper III appears by its title and its place in the catalogue (Tables 2 and 3).
40. The earlier Gemini correlation is printed as +0.311, as Paper IV.d gives it (Section 4.1).
41. "Replicated across multiple scorers" gives way to what the scoring was: the judges scored the same responses, so their agreement is not a count of independent confirmations (Section 4.1).
42. The twelve-domain predictions of Paper VII are described as fixed in public before their fits,

timestamped in advance of them, and never as registered or preregistered (Section 8).

43. The HRIH monograph appears under its title, *Recursive Creation: can recursive self-improvement end in physical creation?*, held apart from the tested claims (Section 11, Table 3 and the references).
44. Wording that offered the author's own work as a suggestion goes: the abstract's description of the $d/(d+1)$ bound goes with the rebuilt abstract, the mechanism of the weight experiment is the candidate mechanism, the Cauchy framework offers a common reason, and the field's status is given in the founding paper's current words (the abstract and Sections 5.2, 12 and 16).
45. Spaced hyphens are kept to a bare minimum, and numeric ranges are written with "to" (throughout; the records keep their printed hyphens).
46. The first-page footnote links the author's Who-is page and the book; the link to Paper C as a credential is removed (the title block).
47. Unit 06 is stated once, as a classification audit that tests no law (Section 14).
48. The HARI Treaty, the Draft Treaty on Hardware-Aligned Recursive Intelligence, is named in full at its first use, the HRIH monograph is held apart from the tested claims, no cosmology leads and no work of fiction is cited (Sections 6.2 and 11, and the references).
49. The drafted preregistration units are no longer called a registered programme of seventy: sixty-five designs are preregistered by timestamp and three study units are registered, each with what it is (Sections 8 and 14).
50. The ARC-Align benchmark of Paper IV.c is described in its own grade, a candidate benchmark whose scorer has not been validated against expert raters, no longer as valid whether or not the framework is correct (Section 10).
51. The code and data are stated as they stand on 9 November 2026, no longer as all published: they are all rights reserved, with one grant, the repository's Replication Licence, version 1.0, under which anyone may download, run, adapt and re-analyse them to reproduce or check the reported results, or to test whether they hold elsewhere, and publish what they find; and the files a run's reproduction notes name as not bundled in the repository are named (Section 10 and the back matter).
52. The reach of the $d/(d+1)$ relation is stated in the words of Paper VII and of On the Origin of Scaling Laws, no longer as matching observed scaling across 50 domains from mice to galaxies (the abstract and Sections 4.3, 9 and 16).
53. Paper VII no longer reports the negative controls of its first run, so the paragraph of Section 4.3 that described them stands as a dated record (Appendix D.4), and the plain-English summary of Section 4.3 no longer says that none matched when the assumptions were violated.
54. The chokepoint is credited to the book and to the HARI Treaty paper, in the book's own words, no longer to Paper I, and the sentence saying that four companies control all advanced semiconductor manufacturing goes (Section 6.2).
55. The labels "Weight v1:" and "Weight v2:", lost in version 2.7, are restored (Section 5.2).
56. "It has not yet been replicated by an independent research group" gives way to "No repetition outside the programme is recorded yet" (Section 4.1).
57. The heading of Section 7.4 becomes "Unblinded results were read as findings": the earlier heading said which direction was correct, which Paper IV.d says the comparison cannot show.
58. No count of corrections is offered as evidence: the catalogue of Section 7 keeps every item and is described as a public record of the programme's practice, not as evidence for the theory (Sections 7, 9 and 16).
59. The causes printed for the null and the inconclusive result are stated as candidate accounts throughout,

as Section 7.12 already did (Sections 5.2, 7.6, 10 and 16).

60. The corrector-class contrast is called the decisive trial, in the words of The ARC Theory, no longer the programme's single most consequential open experiment (Section 6.1).
61. The sentence on a coverage matrix that tied forty-two of the programme's claims to registered tests goes: the tests are drafts, not registered, and no public record of the matrix can be named, so its figure is not printed (Section 8).
62. Phase C's removal test says what a registered verdict of support would support, entanglement under the registered test, by the programme's own run, no longer that entanglement would be demonstrated (Section 8).
63. The spine figure is not reprinted (Appendix A.3).
64. The registration's classification of the gated simulation is cited to version 1.104, whose words it quotes (Section 4.4).
65. The title block's list of integrated documents and its website labels are not set; Section 11 lists every document.
66. Section 7.13 is new: it records, with dates, the corrections found since version 2.7, a review from outside the programme among their sources, each of its leads checked at source before any change, and points back to the entries it refines.
67. Appendix D keeps the dated note, the epistemic-status line, the abstract and the paragraph on the first run's negative controls of version 2.7; version 2.7 itself, with its wording of every passage this version re-sets, rebuilds or cuts, stays in this paper's file history on OSF, and is not reprinted here, so that no corrected wording or withdrawn figure is printed again under this version's DOI.
68. Table 3 lists all 36 documents of the programme's catalogue, generated from the catalogue's data, with the released version of each; the HARI Treaty paper's place is given in that paper's own words, a conditional governance design, rather than in the catalogue's word for a suggestion.
69. The reference to He and Chen (2003), *Fractals* 11, 437, is withdrawn: no article of theirs stands in that journal's volume for 2003, so the derivation it was credited with cannot be read at its source (Sections 7.13 and 12, and the references).
70. Paper V is cited at its version 1.8 and Paper IV.c at its version 1.13, both of 9 November 2026: Paper V's grade of its results is given in its own words, its scoring as that version states it, one model scorer per run, which paraphrased what it scored, and a later five-judge run on DeepSeek V3.2 that its earlier versions left out is reported (Sections 4.2 and 15, Tables 2 and 3, and the references).
71. The blinding layers of the ARC-Align design are given as Paper IV.c specifies them; version 2.7 listed laundered prompts, which that paper does not name among them (Sections 4.1 and 5.3).
72. The sentences that offered the corrections as support for the framework, or praised the programme's own conduct against other programmes, give way to plain statements of the record: the corrections are listed as a record of the programme's practice, not as evidence for its theory (Sections 7 and 16).
73. Results kept from version 2.7 are graded as their own papers grade them, and causes as candidate accounts: the weight experiment's loss curve is what Paper VIII calls a positive signal; the collapse in its removal test is consistent with adapter fragility, which the experiment does not separate from structural entanglement; the DGM null is no longer said to owe nothing to the judge, which was chosen after the earlier judges failed; and the account that the DGM manipulation sat at the wrong level is stated as untested, without praise of the design (Sections 5.1, 5.2, 7.2 and 7.6).
74. The value of α is said to turn on a system's self-referential coupling, with its composition operator constraining only the form through Cauchy's equations, as Paper II states, where version 2.7 said

that it turns on the composition operator (Section 7.1).

75. Stakeholder care is named as scored wherever a result is stated (Sections 5.3 and 7.10).
76. The programme's empirical work is dated from January 2026, when its first papers were published (17 and 22 January), where version 2.7 said February (Section 1.1).
77. The scaling prediction is offered under its premises: a test of the exponent bears on the transport premises, not on Cauchy's equations, where version 2.7 said that a test in any domain could support or falsify the framework; the range of mammalian exponents is accommodated, not predicted, without an independent measure of the effective dimension; and the account of the derivations' convergence is called the programme's own on the search that Paper VII's prior-art record dates (Section 12).
78. The note to the growth solution says that the stable region is bounded, as the paragraph after it sets out, where version 2.7 said bounded below; and a proof of unbounded growth without correction is no longer offered as open work, since the closed form settles it within the cumulative-advantage model, so what remains is to extend the result to wider classes of dynamics (Sections 6.1 and 8).
79. The Darwin Gödel Machine and low-rank adaptation are cited to their sources where the studies first name them (Tables 1 and 2, and Section 5.1).
80. Section 12 is headed "A prediction any researcher can test", Phase A "the nearest tests", and Phase A is said to come first, in place of the business terms version 2.7 used (Sections 8, 9 and 12).
81. Phase C's training run is said to test structural entanglement at frontier scale, no longer to be the definitive test, and Phase A's jailbreak study measures needless refusal of ordinary requests beside resistance (Section 8).
82. Section 4.4 names Paper III and the Foundational paper separately, and says what the drag control was where it is first named: it ran beside the three conditions and matched the static baseline.

A.2 Versions 2.7 to 2.0, as printed

The version history below is kept as version 2.7 printed it, one paragraph for each version. Section and table numbers in it are those of version 2.7, and Table 4 maps them to this edition. Its links now lead to each work's official record.

v2.7 - 5 October 2026. Substantive: the title now follows the programme's uniform order (paper, its law or framework, its own title), *Paper IX · The ARC Theory · Synthesis and Roadmap*, and the title block at the head of the paper carries the same title; until this version it read *Paper IX: Synthesis and Roadmap*. The plate beneath the author line now gives the paper's place in the programme in the words of the programme's map. Mentions of the programme's papers in this paper now carry their uniform titles. Table 3 gives the founding paper at its version 3.4, with its live state in that paper's own words (five laws as named conjectures with statuses declared, where the row had three), and the reference list cites the same version. Section 13 now says that Recursive Dynamics was named as a field at the Carnot stage, where it said proposed. Section 6.2 and Table 3 give Eden Engineering at version 7.0, the version that carries its uniform title, where they gave version 6.3. No registered law, prediction, kill condition or dependency changes; no result is added; no date of any claim moves.

v2.6 - 4 October 2026. Substantive: Section 12 no longer says that the $d/(d+1)$ formula was independently derived by at least seven research groups; as the works read, one is a special case, one could not be confirmed at a registry and the abstracts of two state no exponent formula, so a count of independent derivations granted earlier work more than it stated. The lead now says that the formula is

derived, in its general form or as a special case, in earlier work, and the list gives each work as before. The version number in the title block, the summary and the structured record. No registered law, prediction, kill condition or dependency changes; no result is added; no date of any claim moves.

v2.5 - 4 October 2026. Substantive: the theory gained two laws on 24 September 2026, and the paper now says where it stands against the five. The abstract named three laws as the framework's laws; it now names five, Laws I to III registered on 8 September 2026 and Laws IV and V, added on 24 September 2026, neither registered as a law nor tested, with their mathematics set out at version 1.0 of 3 October 2026. The epistemic-status line beneath the citation block said that every law the programme names is under registered adversarial test; it now says so of Laws I to III and gives Laws IV and V their own status. A dated note before the references places this paper against Laws IV and V. Sections 7.1, 7.8 and 7.12(b), Table 2 and Table 3 printed the interval of −1.3 to 2.9 beside the regression exponent of 0.49, or printed that exponent or the parallel exponent of 0.31 apart from the rest of its row. Section 7.1 and both tables now carry Paper II's row for Gemini 3 Flash whole, as one model's fit and inconclusive, with the interval beside the endpoint exponent of 0.59; Sections 7.8 and 7.12(b) point to Section 7.1. The abstract and Section 8 said that each of the seventy preregistration units was awaiting human submission; they now say that one trial, the corrector-class audit (unit 06), was registered on OSF on 1 October 2026 and has been public since 4 October 2026, and that every other trial is a draft. Table 3 showed Eden Engineering as withdrawn on 14 July 2026; that decision was reversed on 19 September 2026, and the specification stands published at version 6.3. Section 5.3 described the Paper V results as demonstrating stakeholder care as a universal alignment intervention; it now says what they show, an improvement in each of the five analysable runs, model by model. Tables 2 and 3 described Paper II as six-model alignment scaling and listed Claude among its models; they now describe it as Paper II does, capability scaling measured on five models in its tier-2 arm, Claude having been measured only in the alignment arm. The abstract said the work ran between February and August 2026 and reached twenty-six papers; it now says the work began in February 2026 and continues, and that the catalogue holds thirty-two documents. Where the evidence tables said that Paper VIII's gated simulation confirmed the Babylon reward-hacking fingerprint, they now say that it showed it, as Paper VIII now grades that result; where they printed Paper VII's 19/25 alone, they now give beside it the 18/25 of its rerun of 11 August 2026 with the logarithmic candidate, as Paper VII does, without the word confirmed beside it, and Sections 4.3, 9 and 13 give the 18 of 25 beside their 19 of 25; and Section 7 says that each correction illustrates the mechanism the framework describes, where it said demonstrates. Zhao's 2022 derivation is given as from producer-consumer dynamics, as his paper states it, where it was given as from network optimisation. In the same list of Section 12, the paper of Banavar and co-authors of 2010 is given its own result, 2/3 when the velocity of flow is constant and at most 3/4, where it was counted among the derivations of the formula, and Bettencourt (2013) the exponent he derives for a city's land area, $D/(D+H)$, which takes the form only when $H = 1$; the list now counts the formula as derived in its general form or as a special case. The gated simulation of Paper VIII was counted as a positive experiment. The abstract counted it among the empirical studies as a clear positive result, and the plain-English summary beneath it called it positive; Table 1 placed it in a tier of its own, Positive (single experiment), at moderate strength, and Tables 2 and 3 graded it positive; the captions of Tables 1 and 2 and Section 9 counted it as one positive beside one null and one inconclusive result, and the plain-English summary of Table 1 gave it a B; Sections 2, 3, 4.4, 10 and 13 called it the only experiment that produced a clear positive result; the heading of Section 4.4 said that it confirms the Babylon fingerprint, its result paragraph that it confirmed the reward-hacking fingerprint, and Tables 1 and 2 that its drag control was confirmed; Sections 3 and 4.4 said that it succeeded where the other experiments failed, and Sections 3, 4.4 and 13 that the Eden Protocol had produced an effect measurable on the endpoints used at the architectural level. These places now report it as a simulation result, never as

an empirical outcome, and between them give what it showed: the Babylon reward-hacking fingerprint, a gain of 4.5 per cent in capability for a loss of 2.4 per cent in safety, and a drag control that matched the static baseline exactly. Section 4.4, which stands under What Is Proven, now says that the simulation is not among what is proven, and quotes the theory registration's classification of its result; it said that the Eden condition preserved safety, and it now gives the Eden condition's capability and safety scores against the static baseline's, as Paper VIII does. Sections 3, 4.4 and 10 gave the design of the simulation, deterministic metrics on a system without pre-existing RLHF training, as the reason for its result; they now give that as the candidate account Section 7.12(c) records, and the abstract, which called it the one experiment in the programme of that design, now calls it the one Paper VIII experiment of that design. The plain-English summary of Section 4.4 gave the source of the cost of safety as a general fact; it now gives it as what the simulation showed. Table 2 graded the simulation of Paper VI as supported, and Tables 2 and 3 said that entangled safety prevents collapse across twenty adversarial seeds; both now report it as a simulation result and give what Paper VI reports: collapse averted in a single-seed run and, across the twenty seeds, a separation on safety retention and none on collapse. Section 7.12(c) refers to the account of why the gated simulation showed the pattern, where it said why it succeeded. The abstract gave the resistance of RLHF-trained models to prompt-level differentiation and the strength of instruct tuning against LoRA as the reasons for the null and the inconclusive result; it now gives each as the candidate account Section 7.12(c) records. It said that every remaining study gave statistically significant findings under blinded evaluation; it now gives what each paper reports: such findings in Papers IV.a to IV.d; in Paper II, a parallel exponent at or close to zero in four of five models, an earlier estimate that did not replicate, and a sequential exponent that comes from the fit of one model and is inconclusive; in Paper V, significant gains scored across models and not under the four-layer blinding protocol; and in Paper VII, a match that is significant under a permutation test, in a structured prediction comparison and not a blind trial. Section 6.2 gave hardware-level embedding a readiness level of TRL 0-1, with no prototype, and a programme of five to ten years; it now gives what version 6.3 of Eden Engineering prints: four mechanisms graded separately, from TRL 4-5 to TRL 1, with timelines from 1-2 years to 10+ years, and a falsification condition recorded as untested and requiring a hardware prototype. Editorial: the abstract called Recursive Dynamics a proposed field, and Section 1.2 and Table 3 called its founding paper the field proposal; the abstract now names Recursive Dynamics as the field the author founded, and Section 1.2 and Table 3 call that paper its founding paper. Section 13 keeps the status the founding paper itself gives the field: proposed, at the Carnot stage. The plain-English summary of Section 7 dates the five corrections of Section 7.12 to version 2.4, and Section 4.3 no longer names the version of Paper VII that carries the permutation test. The reference entries for Papers I and IV.c, Eden Engineering, the Darwin Gödel Machine and Zhao (2022) print the titles, authors and identifier their official records give; the entry Demetrius (2010) named an article that does not exist and now gives the 2008 book chapter that carries its title; the entry He and Chen (2003) names the one record of theirs that could be found; the entry Sandberg and Bostrom (2008) says that no official online copy is available; cited works are linked to their official sources where one could be confirmed, and the programme's papers to their DOIs; the book and the two sources named in the copyright notice are linked; the page's title, dateline, citation records and structured record carry the new version and date. No registered law, prediction, kill condition or dependency changes; no result is added; no date of any claim moves. First published 18 March 2026; that date is unchanged.

v2.4 - 20 September 2026. Substantive: five corrections are added as Section 7.12, and the correction tally moves from fourteen to nineteen. The displayed solution in Section 6.1 is restricted to the open leverage interval, with the exponential branch at one and the finite-depth branch above it named. Section 7.1 returns the wide interval to the retracted estimate it belongs to and prints the later estimate's own

interval, which settles neither the sign, nor the regime, nor the comparison with the bound. The null result is recorded as no difference detected on the endpoints measured rather than as identity, and the root causes standing beside the null and the inconclusive result become candidate accounts with rival accounts left open. Section 4.3 names which null the proportion comparison is taken against and holds to the permutation as primary. The paragraphs that depend on Papers V, VI and VIII point to the corrected passages instead of restating them, and the Paper V phrasing is scoped to the five runs that could be scored, of six run. Three proposed repairs were refused and the refusals are recorded with the corrections. Also substantive: the preregistration estate is stated at seventy, the number the register carried when it was last generated; an earlier entry in this block prints seventy-four as at 1 September 2026, and that entry stays as written because it records what v2.1 set, seventy being the register's figure; the founding paper's conceded objections read thirty-nine, correcting the thirty-eight printed at v2.3; the convergence register's rows read thirty-three; Paper XII shows at protocol stage rather than as a registered protocol, the monograph's prediction ledger as dated rather than registered, and the monograph as speculative; and reviews run inside the programme are no longer called independent, in Section 7.9, in the Paper XI row of Table 3 and in the reference title for Paper VIII. Editorial: Paper VII is cited at v3.11 and the Recursive Dynamics founding paper at v2.8 in both the map and the reference list; the retracted point estimate carries the word retracted in the heading and first sentence of Section 7.8; the two table captions take this version's date; the served comment above Section 3 matches its heading; and an instruction to an editor is removed from the body of Section 7.1. No result, no reported figure and no status has been recomputed, and nothing is submitted anywhere. First published 18 March 2026; that date is unchanged.

v2.3: The founding paper's objections read fifty-one, thirty-nine of them conceded in full or in part, after one objection was withdrawn on 16 September 2026. The figure of thirty-eight printed at v2.3 was wrong and is corrected here. No claim, date, result or status has changed.

v2.2: The provenance note at the end has been corrected for accuracy. No claim, date, result or status has changed.

Paper IX v2.1 - 2 September 2026: the Paper VIII tally now reads one null, one inconclusive and one positive across three experiments, where the text had claimed two nulls across four; Paper V leaves the proven row of Table 1 for a supported one, in step with Section 4.2, and its five analysable runs and three loops now carry the names Paper V gives them, the superseded text having listed GPT among the five and having described loops Paper V does not test; Paper VII's headline is the permutation test, with the binomial standing beside it as a sensitivity check, and the negative controls are attributed to the v1 run and its record; the account of scoring is put right, a purpose-built Claude Sonnet scorer adapter having scored the Claude Opus run; the self-reference exponent takes the subscript β_{L} that the notation spine sets; the measured exponent appears with its interval, and the parallel exponent as four of five models; the preregistration estate stands at seventy-four (as at 1 September 2026); the window in which the programme was produced runs to the present; Table 1 carries this version's date; Eden Engineering shows as withdrawn in Table 3 and in the footer; and the reference titles for Papers IV.a, IV.b and VII are corrected.

v2.0 - 1 September 2026: a structural refresh drawing in Papers X to XIII, HRIH, Paper C and the Recursive Dynamics founding paper.

A.3 The figure of version 2.7

Version 2.7 printed a figure after Section 7.12, its caption "The spine, drawn: every claim mapped to its source and the condition that would kill it. Source: research-evidence-spine.json." Its artwork, an inline vector graphic, showed an eighteen-claim register in four groups (original, replicate, speculative, shared), an interpretation-guard panel, a panel headed "Grant relevance", and a passport line dating Paper IX to 2 January 2026. It is not reprinted: the catalogue dates Paper IX to 18 March 2026, the grades are a register's and not this paper's, and the panel and labels are website material. The caption above is quoted as printed; the figure stands in version 2.7's PDF, which stays in this paper's file history on OSF.

A.4 Where the parts of version 2.7 stand

Table 4. Where each part of version 2.7 stands in this edition. Section numbers on the left are version 2.7's, and version 2.7's own wording of each part stays in that version's file on OSF. Table 3 is generated from the programme's catalogue data as they stood on 7 November 2026.

Part of version 2.7	Where it stands in this edition
Title, author line, plate and subtitle, with the version line and the list of documents it integrated	Title block and first-page footnote; the earlier title, subtitle and lines in Appendix A.1
Abstract	Abstract (new); version 2.7's in Appendix D.3
The box beneath the abstract	Its pointer in Section 13
What This Paper Shows, in Plain English	The opening of Section 1
1. Introduction	Section 1
1.1 Timeline and Priority	Section 1.1
1.2 Scope	Section 1.2; two of its paragraphs folded into the opening of Section 1
2. The Evidence Hierarchy, with Table 1	Section 2, with Table 1 (rebuilt)
3. The Programme's Empirical Studies, with Table 2	Section 3, with Table 2 (rebuilt); its box on the gated simulation not set
4. What Is Proven	Section 4, What has been shown
4.1 Alignment Scaling Is Architecture-Dependent (Papers IV.a-d)	Section 4.1
4.2 Stakeholder Care Is the Most Robust Intervention Signal (Paper V)	Section 4.2
4.3 Cauchy-Predicted Scaling Families Match Empirical Data (Paper VII)	Section 4.3; its paragraph on the negative controls in Appendix D.4
4.4 Gated Simulation Shows Babylon Fingerprint (Paper VIII, Experiment 3)	Section 4.4
(new)	Section 4.5, the first registered verdict
5. What Produced Null or Inconclusive Results	Section 5
5.1 DGM Null Result at the Prompt Level (Paper VIII, Experiment 1)	Section 5.1
5.2 Weight-Level Structural Entanglement (Paper VIII, Experiment 2)	Section 5.2

Part of version 2.7	**Where it stands in this edition**
5.3 Blind Replication of Eden Intervention (Paper V Gap)	Section 5.3
6. What Is Theoretical (Untested)	Section 6
6.1 Unbounded Scaling Under Recursive Self-Modification	Section 6.1
6.2 Hardware-Level Embedding	Section 6.2
7. What the Programme Got Wrong	Section 7
7.1 to 7.3, 7.5 and 7.7 to 7.12, each under its heading	Sections 7.1 to 7.3, 7.5 and 7.7 to 7.12, under the same numbers
7.4 Unblinded Evaluation Produced Misleading Results	Section 7.4, Unblinded results were read as findings
7.6 DGM Experiments at the Prompt Level Cannot Test the Hypothesis	Section 7.6, DGM experiments at the prompt level may not have tested the hypothesis
The figure after 7.12	Not reprinted; Appendix A.3
(new)	Section 7.13, the corrections since version 2.7
8. The Phased Roadmap	Section 8
9. For Funders	Section 9
10. For Peer Reviewers	Section 10
11. The Complete Programme Map, with Table 3	Section 11, with Table 3 (generated from the catalogue's data)
12. The Pitch to Researchers	Section 12, A prediction any researcher can test
(new)	Sections 13, 14 and 15
13. Conclusion	Section 16
This paper and Laws IV and V, 3 October 2026 (a note at the end of Section 13)	Appendix D.1
References	References
The version history	Appendix A.2
How to cite this paper	Appendix E
The epistemic-status line	Section 14; version 2.7's line in Appendix D.2
The copyright line	The back matter; an account of version 2.7's line in Appendix D.2
The standing covenant	The back matter
The provenance note	Acknowledgements
(new)	Appendices B and C

B Priority record

Each dated record below carries its kind of timestamp: a time signed by mail servers, a Bitcoin anchor of a design's hash, a deposit dated by its server, or a date written inside a document. No kind is presented as

stronger than it is.

Table 5. The priority record: each dated item, where it stands and its kind of timestamp.

Date	Record	Where it stands	Kind of timestamp
8 December 2024, 02:45:18 UTC	The self-emailed manuscript: the ARC Principle, the embedded-alignment thesis it names the Eden Protocol, and the term recursive dynamics	Header block and check script on OSF (osf.io/qasrk) (Eastwood 2026j)	Times signed by the mail servers: the Date field covered by the sender's DKIM signature (signed 02:45:30 UTC) and Google's ARC set (sealed 02:45:31 UTC), verified on 8 October 2026 against keys published in DNS; the body's hash fixes its content.
2 January 2026	*Infinite Architects*, first edition (print; ebook 6 January 2026)	ISBN 978-1806056200; osf.io/dqm3e (Eastwood 2026a)	Publication date of the printed book (ISBN record); the first edition deposited on OSF (server-dated).
18 March 2026	Paper IX first published	osf.io/k7ruz (node created 20 March 2026, 17:03:52 UTC)	The paper's own first-publication date (document date, the catalogue's); OSF server date of the node.
12 August 2026	The first designs preregistered by timestamp (58 of the 65; 7 more on 28 August 2026 at block 964465)	Bitcoin block 962066	Bitcoin anchor of each design's hash.
8 September 2026	The theory registration, version 1.100: twenty-two propositions; Laws I to III	DOI 10.17605/OSF.IO/P8CKQ (Eastwood 2026ah)	OSF server (registration accepted).
24 September 2026	Laws IV and V added to the theory	The ARC Theory (statement paper), dated note (Eastwood 2026ai)	Date written in the document.
15 October 2026	Laws IV and V registered as laws (version 1.103)	DOI 10.17605/OSF.IO/P8CKQ	OSF server (approved update, submitted 11:14:28 UTC).
1, 22 and 27 October 2026	Units 06, 83 and 82 registered	DOIs RV2B5, MN73W, JUFG4 (Eastwood 2026ak,an,am)	OSF server (registrations accepted).
29 October 2026	Version 1.104 of the theory registration; Unit 83's results	DOIs P8CKQ and 98HKR (Eastwood 2026ao)	OSF server.

C Notation and glossary

Table 6. Notation. Each symbol is defined where it is first used; the third column gives the place.

Symbol	Meaning	Where
α	Law I's exponent of recursive depth, measured rather than assumed; in the cumulative-advantage model $\alpha = 1/(1-\beta_L)$, above one for $\beta_L \in (0,1)$	Sections 4.3, 6.1 and 7

Symbol	Meaning	Where
α (Section 12)	The asymptotic throughput-cost exponent of a declared network, $\mu/(\mu+\zeta)$, which equals $d/(d+1)$ under the transport premises	Section 12
β_L	Self-referential coupling in the cumulative-advantage model, $dg/dr = ag^{\beta_L}$	Section 6.1
g, r, a	Capability, recursive depth and rate coefficient in that model	Section 6.1
γ	Correction-leverage exponent; never measured	Sections 6.1 and 7.11
α_{crit}	Law III's stability ceiling, $1/(1-\gamma)$; the ARC Bound, at most two, is its value at $\gamma = 1/2$	Section 6.1
β, k	Paper X: exponents of correction strength and of the specific growth rate in capability; Law II's criterion is $\beta > k$	Section 6.1
δ	Paper XIII: exponent of cycle duration, $\Delta t \propto C^{-\delta}$	Section 6.1
C	Capability (Papers X and XIII); in Section 12, flow-length cost	Sections 6.1 and 12
d	Two uses, kept apart by context: the dimension in $d/(d+1)$; Cohen's d, a standardised effect size	Sections 4 and 12
μ, ζ, L, B, $\bar{\ell}_q$	Demand exponent, mean-path exponent, linear size, demand and mean path length of a declared network	Section 12
U, I, R, f	The equation line of Section 4.3: usable output, base quantity, recursive depth and the form set by composition; there β is the self-referential coupling, written β_L above, and k the rate of the exponential form, not Paper X's symbols of the same letters	Section 4.3
$\mathcal{L}_{\text{cap}}$, $\mathcal{L}_{\text{safe}}$	Capability and safety losses of Paper VIII's entangled loss	Section 5.2
ρ	Spearman's rank correlation	Section 4.1
p	The probability, if there were no effect, of a result at least as extreme as the one observed	throughout
P1 to P22	The twenty-two propositions of the theory registration	Sections 4.5 and 14

The terms below are defined in the words of the theory registration where it defines them, and otherwise in the words of the paper that does (Eastwood 2026ah, version 1.104).

law under test A law that no registered test has yet supported or refuted as a law. The registration fixes the words a law can take: "A law has three words: supported, refuted and under test. Under test is printed with the mark of each proposition the law's rule names and with what is missing" (Eastwood 2026ah, version 1.104). Every law of the theory is a law under test (Section 14).

registered Submitted to OSF by the author and accepted there (Eastwood 2026ai). A registered proposition on which nothing has yet been observed stands at the lowest rung of the evidence ladder, whose permitted wording is "registered, with the identifier stated, and not tested" (Eastwood 2026ah, version 1.104).

preregistered by timestamp Of a design: "the text of a design is anchored in the Bitcoin blockchain at a stated date" (Eastwood 2026ai). A timestamp dates a design; it does not register it.

precursor evidence Evidence from a run made before any registered test of its question (Section 1.2).

Of runs made before the theory registration, the registration's words are that such evidence is "not confirmatory under any condition this document registers", and it decides no proposition there and is scored against none (Eastwood 2026ah, version 1.104).

prediction-side In the registration's words: "any simulation output, any design-sensitivity rate, and any value a model returns rather than an instrument measures is a prediction-side artefact, is reported as one wherever it appears, and is never counted, tabulated or described as an empirical outcome" (Eastwood 2026ah, version 1.104).

administratively incomplete The standing of a registered study whose registered run could not be completed under its failure policy: it is reported in full and, in the registration's words, "as administratively incomplete under its registration, with no inferential verdict" (Eastwood 2026ah, version 1.104). Section 14 gives the one such study.

candidate account An explanation offered for a result and not tested against its rivals. Section 7.12, item (c), records the causes printed beside the null and the inconclusive result, and the account of why the gated simulation showed its pattern, as "candidate accounts rather than findings", with rival accounts unexcluded.

rung (of the evidence ladder) One of the eight levels the registration fixes for the evidence a proposition has reached, each with "the exact public wording it permits"; "No surface of this programme may use wording from a rung above the one a proposition has actually reached" (Eastwood 2026ah, version 1.104). A registered verdict from the programme's own run stands at rung 1 (Section 4.5).

judged score A score that a model judge gives a response against a rubric. The registration sets it apart from a count, a term that is "a count settled by frozen code against ground truth fixed before the run" (Eastwood 2026ah, version 1.104).

Static, Babylon and Eden conditions The selection conditions of the self-improving agent experiment of Paper VIII, in its words: Static, "Control condition with no mutation"; Babylon, "Capability-only selection"; and Eden, "Entangled $C \times S$ selection with a safety floor and reward hack detection" (Eastwood 2026ab). The gated simulation used conditions of the same names, with a drag control beside them (Section 4.4).

DGM (Darwin Gödel Machine) The system of Zhang et al. (2025), in their words "a self-improving system that iteratively modifies its own code (thereby also improving its ability to modify its own codebase) and empirically validates each change using coding benchmarks". The first experiment of Paper VIII, DGM v3, follows that framework: it evolved agents by code-level self-modification under the three conditions above, with a judge model, GPT-5.4, scoring them (Section 5.1).

ARC set (of an email) A set of assertions under the Authenticated Received Chain protocol, by which each entity that handles a message can, in its specification's words, "attach assertions of message authentication assessment" to it (Andersen et al. 2019). Google's ARC set on the record of 8 December 2024 is such a seal (Section 7.3); the protocol is unrelated to the ARC Theory.

access axis and mechanism axis (Unit 83) The two readings of P21's registered verdict. The access axis is what the deciding unit varies: in the registration's words, "what it manipulates is information access, whether the corrector sees the process or only its output", and P21 "is therefore decided on the access axis". The mechanism axis "asks whether the monitor's detection rate falls as the artefact it inspects grows", and "whether the part of the widening that fall can carry clears the tolerance once the part owed to the outside arm's action space is separated out"; it is reported beside the verdict and never in its place (Eastwood 2026ah, version 1.104). In Unit 83 it was not evaluable (Section 4.5).

D Dated notes and records of version 2.7, as printed

This appendix keeps, as version 2.7 printed them, its dated note of 3 October 2026, its epistemic-status line, its abstract and its paragraph on the negative controls of the first run of Paper VII, with an account of its copyright line. Every other passage of version 2.7 stands in that version's file, which stays in this paper's file history on OSF (DOI 10.17605/OSF.IO/K7RUZ), and Table 4 says where each part of it now stands. Section numbers inside these records are version 2.7's. Their links now lead to each work's official record, and square brackets mark an editorial addition.

D.1 The note of 3 October 2026

This paper and Laws IV and V, 3 October 2026. The theory has been stated with five laws since 24 September 2026. Law IV, the ARC Persistence Law, is carried by proposition P15, and Law V, the ARC Embedding Law, by propositions P12 and P21. The three propositions are registered; neither law is registered as a law, and neither has been tested. Their mathematics is set out in Laws IV and V: the mathematics, version 1.0, of 3 October 2026. This paper synthesises the programme's studies and their corrections. Its summaries carry the limits of the experiments they summarise, and they add no evidence of their own for Law IV or Law V. No result in this paper is a deciding test of P15, P12 or P21. Naming the two laws changes nothing printed here: the results, their dates and their limits stand as they were.

[Section 14 gives the status of Laws IV and V, and of their mathematics, as it now stands.]

D.2 The epistemic-status line and the copyright line of version 2.7

Epistemic status. What this programme names Laws are conjectures. Laws I to III are under registered adversarial test. Laws IV and V were added on 24 September 2026; the predictions that carry them are registered, the laws themselves are not yet registered, and neither has been tested. Every quantity in this paper is operationally defined, and established-law standing is claimed nowhere. The registered programme exists to earn that standing, or lose it, by measurement, replication and survived refutation.

[Section 14 gives each law's status as it now stands.]

Version 2.7's copyright line asserted full human authorship and moral rights under the Copyright, Designs and Patents Act 1988; the clause that followed it, its clause on technical contributions and its link to a fuller statement are not set here (version 2.7 prints them). The back matter of this version carries its copyright line.

D.3 The abstract of version 2.7

Abstract

The ARC/Eden research programme now spans a full set of documents and a run of separate empirical studies covering mathematical foundations, methodology, empirical validation, engineering specification, and philosophical vision. The core concepts were first articulated on 8 December 2024 in an email bearing a visible Gmail origination field, with sender-DKIM and paired Google ARC mathematics verifying against keys captured at the signed selectors and no RFC 3161 trusted timestamp. The book *Infinite Architects* (ISBN 978-1806056200) was published 2 January 2026 (print; ebook 6 January). Work on the research programme began in February 2026 and continues. Its output now reaches the thirty-two documents of its catalogue; a registered programme of seventy drafted preregistration units, the number the register carried when it was last generated on 17 August 2026, each carrying a schedule drawn from a published seed, whose regeneration is checked by the estate's own tools rather

than asserted here; and Recursive Dynamics, the field the author founded, whose founding paper states the kill condition for its own name (DOI 10.17605/OSF.IO/HCPBU). Of the trials in the programme, one, the corrector-class audit (unit 06), was registered on OSF on 1 October 2026 and has been public since 4 October 2026 (DOI 10.17605/OSF.IO/RV2B5). Every other trial is a draft awaiting submission by the author, most of them preregistered by timestamp. This synthesis paper integrates all findings into a single honest assessment. The framework's laws, five since 24 September 2026, are carried as named conjectures with the status printed beside the name: Law I, the ARC Principle; Law II, the ARC Co-Scaling Law; Law III, the ARC Ceiling, whose ARC Bound of two is the Ceiling's value at one half; these three were registered on 8 September 2026. Law IV, the ARC Persistence Law, and Law V, the ARC Embedding Law, were added to the theory on 24 September 2026; neither is registered as a law or tested, and their mathematics is set out at version 1.0 of 3 October 2026. The correction leverage gamma has never been measured, by anyone. The empirical studies break down like this. Paper VIII returned a null result in 1 case, DGM v3, on the candidate account that RLHF-trained models resist prompt-level differentiation, and inconclusive results in 1 case, weight-level LoRA fine-tuning, on the candidate account that instruct tuning was too strong for LoRA to override at the ranks tried. Papers IV.a to IV.d report statistically significant findings under blinded evaluation. In Paper II, also blinded, four of the five models it measured put the parallel exponent at or close to zero; an earlier super-linear estimate did not replicate and is retracted, and the sequential exponent now reported comes from the fit of one model and is inconclusive. Paper V reports statistically significant gains in stakeholder care in each of its five analysable runs, scored across models and not under the four-layer blinding protocol of Paper IV.c. Paper VII reports a match between predicted and observed scaling families that is statistically significant under a permutation test, in a structured prediction comparison and not a blind trial. Simulations are not counted among the empirical studies; they are reported as simulation results. The gated simulation, the one Paper VIII experiment that uses deterministic metrics on a toy system without pre-existing RLHF training, showed the Babylon reward-hacking fingerprint: unconstrained optimisation traded safety for capability. The $d/(d+1)$ geometric scaling bound, proposed as an invariant relation under test, matches observed scaling across 50 domains from mice to galaxies, and Cauchy functional equations fix the functional form once the composition identity and a regularity condition are given, and saturation is admitted as a fourth, capped case rather than as a solution of those equations. This paper maps exactly what has been proven, what has not, and what experiments would resolve each open question. It also documents the programme's errors and corrections, because a framework built on iterative self-correction must practise what it preaches.

Keywords: AI alignment, synthesis, evidence hierarchy, ARC Principle, Eden Protocol, Cauchy scaling, self-modifying AI, developmental alignment, structural entanglement, research roadmap

[The abstract of this version replaces it, and Section 7.13 lists the corrections made to its statements since version 2.7.]

D.4 The negative controls of the first run of Paper VII

Section 4.3, a paragraph. Not set in this edition's text: Paper VII no longer reports the negative controls it describes, so it stands here as a dated record of the first run (Section 7.13, item (xx)). Version 2.7 read:

Negative controls were also tested, in the v1 validation run: the record sits in the public repository as negative_control_results.json, 17 March 2026, and Paper VII now carries the permutation test where those controls once stood. Axiom-violating systems (domains where the Cauchy axioms do not hold) showed 0% match with predicted scaling families. Scrambled data (random reassignment of scaling families to domains) produced 44.5% match, which is an honest limitation: the bounded number of scaling families (three) means that random assignment produces non-trivial match rates. The significance rests on the permutation, which conditions on the observed family margins and so already absorbs that scramble rate. Where the observed 76% is set against an expected 33%, that 33% is the uniform three-family null belonging to the binomial sensitivity check and not the scramble baseline; the two are different nulls and cannot carry one comparison between them (Section 7.12(d)).

E How to cite this paper

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BibTeX:

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@misc{eastwood2026paperIX,
  author = {Michael Darius Eastwood},
  title  = {Synthesis and Roadmap: which of the programme's claims
            survived their tests, and which must be tested next?},
  year   = {2026},
  note   = {Version 2.8; first published 18 March 2026},
  doi    = {10.17605/OSF.IO/K7RUZ}
}
```

A citation of an earlier version keeps the title that version printed.